What the Model Learns from Coded Audio: Null-Calibrated Feature Attribution in Explainable Speech Emotion Recognition

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Abstract—Speech emotion recognition systems are validated on uncompressed audio and then deployed on coded audio. Earlier work has asked whether that gap matters for accuracy. This paper asks whether it matters for the evidence a model uses, and it uses models whose evidence can be read exactly. Fifteen classifiers and a small neural network were fitted to 436 named acoustic descriptors extracted from CREMA-D. Every clip was rendered in four conditions derived from a single canonical reference: uncompressed WAV, MP3 at **64** $\mathrm{kbit\,s^{-1}}$, MP4/AAC at **64** $\mathrm{kbit\,s^{-1}}$, and an AAC re-wrap that holds the codec output fixed while changing only the container. We report three results. The first result is that the severe loss observed across formats is caused by train/serve mismatch and by nothing else. A kernel machine trained on WAV scores 0.203 balanced accuracy on MP3, but the same model trained on MP3 scores 0.589 against 0.586 on clean audio. The affective information therefore survives the codec intact. The second result is that what the model learns from that information does change, and the change is readable because the features are named. An entropy tree fitted separately on each condition selects the same root descriptor (prosody_intensity_std) in all four conditions, with information gain stable to within 0.003 bits. AAC then moves the threshold by 1.3 % and replaces the descriptor at depth 1, while the MP3 tree does not. That agreement does not reach far. Grown to depth 6, the MP3 tree reproduces the reference structure to depth 2 and the AAC tree only to depth 0, against depth 3 for an informationally empty container control. The third result is methodological. A codec effect on a feature-importance ranking cannot be interpreted without a null, because the ranking itself is fragile. Refitting on identical audio with different tie-breaking retains 24.2 of 25 top features, but an informationally empty container re-wrap already costs 8.4 of them. Measured against that floor, the codec effects are real but much smaller than they first appear. MP3 retains 12.6 top features and AAC retains 10.7 ($p \leq 1.1 \times 10^{-3}$, paired within resampled training sets). Roughly three fifths of the apparent displacement is therefore generic sensitivity to perturbation rather than a codec-specific effect. We report the complete 436-column inventory and the full per-condition split structure, and we argue that attribution-shift claims in affective computing should require a matched empty-perturbation control as standard.

Index Terms—Speech emotion recognition, explainable artificial intelligence, perceptual audio coding, feature attribution, decision trees, information gain, covariate shift, null calibration, affective computing.

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Code, stimulus manifests, the feature table schema, and every result file underlying the tables and figures are released with this paper. See the Data Availability Statement.

I. Introduction

AUDIO almost never reaches an inference endpoint in the same representation in which it was captured. Contact-centre archives hold MP3. Streaming clients negotiate Opus or AAC. Broadcast pipelines resample and requantise. Evaluation practice runs the other way, because benchmarks are distributed as uncompressed WAV and are read directly from disk.

This gap is usually dismissed on a reasonable argument. Any ingestion path decodes the audio to linear PCM before features are extracted, so the delivery format looks like an implementation detail. Where the argument has been tested, it has been tested on accuracy, and the answers are mixed. Reddy and Vijayarajan [1] report codec effects on speech emotion recognition (SER) that are real, modest, and dependent on the front end.

Accuracy is the wrong instrument for a system that is increasingly required to report both what it concluded and why. A deployed affective system may present an attribution over acoustic evidence to a clinician, an auditor, or a quality-assurance analyst. That second output has its own robustness properties, and those properties do not follow from the accuracy of the first. Ghorbani et al. [2] established for image classifiers that inputs which are indistinguishable to a human observer, and which receive identical predictions, can still be assigned very different attributions. Alvarez-Melis and Jaakkola [3] showed the same fragility in a more general setting. Both studies used perturbations that were constructed adversarially. A transcode is not adversarial. It is routine, it is everywhere, and it is almost never logged.

A. Why interpretable models, and why named features

Answering a question about evidence requires a model whose evidence can be inspected, and inspectability is not a single property. An end-to-end audio model learns its own representation, so an attribution placed on one of its latent dimensions is not a claim about anything an acoustician would recognise. A model fitted to named descriptors in the GeMAPS [4] and openSMILE [5] tradition behaves differently. An attribution on prosody_cpps is a claim about breathiness, and a decision-tree split on spectral_contrast_6_mean is a claim about the 6 kHz to 8 kHz band. This paper is confined to that regime deliberately. Everything reported here is exact rather than estimated, it can be checked against an acoustic reading, and it can be reproduced from the released artefacts.

B. The problem with attribution-shift claims

A claim of the form "compression changed which features the model relies on" is easy to produce and hard to justify. Feature-importance rankings are known to be unstable. Two descriptors that split the data almost equally well can trade places, and a single swap near the root of a tree reroutes every node below it. A ranking can therefore move a long way even when the data has not moved at all. The literature already contains the general warning. Adebayo et al. [6] show that some saliency methods are independent of the model they claim to explain, and Krishna et al. [7] show that different methods disagree on the same fitted model. The specific discipline that this warning implies, which is to measure the shift induced by a manipulation that carries no information, is not yet standard practice.

We adopt that discipline here, and it changes the answer. Our design supplies a natural empty perturbation, which is the same codec output routed through two different containers. Once that floor is measured, most of the apparent codec effect on importance rankings turns out to be generic sensitivity to any input perturbation.

C. Contributions

1) A matched-format decomposition of the codec effect. Training and serving in the same coded condition recovers almost all of a severe cross-format loss, which reaches 38.6 balanced-accuracy points. This establishes that $64\,\mathrm{kbit\,s^{-1}}$ coding does not remove the affective information. It moves that information out of the range the model was fitted on (Section VI-C).
2) Per-condition tree structure as a readable attribution. An entropy tree fitted independently on each condition shows exactly which test the codec changes and which test it leaves alone, at the level of a named descriptor and a numeric threshold (Section VI-D). A depth profile then establishes how far down that agreement reaches, which is two levels for MP3 and none for AAC against three levels for an informationally empty control (Section VI-E).
3) Null calibration for feature-importance shift. We measure three floors, namely estimator tie-breaking, actor resampling, and an informationally empty container re-wrap. A paired within-sample design then separates the codec effect from all three of them (Section VI-F).
4) A repair that is both diagnosable and cheap. Neutralising a single named descriptor recovers 98 % of a kernel machine's loss and 92 % of a neural network's loss, and it costs nothing on uncompressed audio (Section VI-I).
5) Two negative results that we report rather than smooth over. Post-hoc attribution methods disagree substantially on the same fitted model, and a global attribution ranking can report that nothing has changed while the classifier has in fact stopped working (Sections VI-G2, VI-H1).
6) A complete inventory, released with the paper. All 436 descriptors are defined and enumerated (Appendix A), and all 31 internal nodes to depth 4 of every per-condition tree are tabulated (Appendix B).

II. Related Work

A. Perceptual coding and speech task degradation

The degradation of speech systems under lossy coding is documented, but it is not characterised in much depth. Reddy and Vijayarajan [1] find that standard codecs reduce SER accuracy by an amount that depends on both the codec family and the feature representation. That result already implies that coding interacts with the front end rather than degrading performance uniformly.

A result from neural codec benchmarking carries more methodological weight. Wu et al. [8] demonstrate that signal-domain fidelity metrics fail to predict whether task-relevant information is retained. A codec can score well on reconstruction error while discarding the paralinguistic content that downstream tasks require. We take this finding seriously enough to report signal-domain displacement and model behaviour as two separate quantities throughout the paper, and we treat neither one as a substitute for the other.

B. Interpretability for tabular and audio models

Post-hoc attribution has converged on a small set of standard tools. These are local surrogate fitting [9], Shapley-value attribution [10] together with an exact polynomial-time algorithm for tree ensembles [11], and permutation importance. For differentiable models, path-integral methods [12] and reference-based decompositions [13] are standard, and consolidated implementations are available [14]. Broad surveys map the field [15]. Rudin [16] argues that for high-stakes decisions the post-hoc layer should be replaced by an intrinsically interpretable model. Our result in Section VI-A speaks to that position directly, and it does not support it.

Whether these methods transfer to audio is contested. Haunschmid et al. [17] argue that audio explanations should satisfy a listenability criterion, and that transplanting image-segmentation logic onto time-frequency representations produces components with no perceptual correlate. Related formulations appear in [18] and [19]. Nasr et al. [20] extend the critique to speech emotion specifically, and observe that a salient time-frequency region is not for that reason an acoustically meaningful one. Attributing over named acoustic descriptors avoids this objection rather than answering it. Every unit of attribution in this paper has a phonetic reading by construction, and the cost of that guarantee is that the model is confined to hand-designed features.

C. Attribution stability as a validity requirement

The literature that bears most directly on this paper concerns the fragility of attribution itself. Ghorbani et al. [2] show that attributions can be altered substantially by perturbations that fall below the threshold of human perception, while predictions and confidences remain stable. Adebayo et al. [6] demonstrate that certain widely deployed saliency methods are statistically independent of both the model parameters and the data-generating process. Those methods produce outputs that have the surface form of an explanation and none of its content. Krishna et al. [7] document that different explanation methods routinely disagree on the same model, and that practitioners resolve the disagreement by means that have no principled basis.

Taken together, these results imply that attribution stability must be measured against a null rather than assumed. Section VI-F puts that requirement into practice for the specific object at issue here, which is a feature-importance ranking. The requirement turns out not to be a formality, because applying it changes the size of the reported effect by a factor of roughly two and a half.

D. Distribution shift

The mechanism we identify is a textbook case of covariate shift [21], with one unusual property. The shifted variable is not a proxy for anything that the label depends on, because masking it costs nothing on clean audio (Section VI-I). The shift is therefore pure contamination rather than a change in the predictive relationship, and that is why the repair turns out to be so cheap.

E. Where the gap lies

Coding effects have been characterised for accuracy but not for evidence. Attribution fragility has been characterised for vision under adversarial perturbation but not for audio under the routine degradation that occurs in practice. We identify no prior work that isolates the container under a verified constraint of signal invariance, and none that calibrates a feature-importance shift against a matched empty perturbation.

III. Corpus, Stimuli, and Conditions

A. Corpus

CREMA-D [22] comprises 7,442 clips from 91 actors, each delivering one of twelve fixed sentences in one of six acted emotions (anger, disgust, fear, happiness, neutral, sadness), with crowd ratings collected separately for voice-only, face-only and audio-visual presentation.

The fixed set of sentences carries real methodological weight. Holding the words constant across emotions and conditions removes the lexical channel as a source of discriminative variance. Any signal the models recover must therefore be carried by acoustic realisation, and acoustic realisation is exactly what perceptual coding operates on.

One figure is worth stating here for reference throughout the paper. CREMA-D's own crowd raters, hearing the audio alone, matched the actor's intended emotion on only 45.5 % of clips, with per-class recall ranging from 0.966 for neutral down to 0.164 for sad. Models trained on the intended label can exceed that figure because they exploit systematic acting cues that human listeners do not consciously decode. We report the figure as context for the absolute accuracies below, and not as a target.

B. Reference construction and conditions

The canonical reference is 16 kHz monophonic 16-bit linear PCM, loudness-normalised to EBU R128 ($I = -23\,\mathrm{LUFS}$, $\mathrm{LRA} = 7$, $\mathrm{TP} = -2\,\mathrm{dB}$). Coded conditions are produced from that reference by a single encoder invocation each, and all decoding passes through one identical FFmpeg call so that the decoder never confounds the codec.

The fourth condition is the device that lets us identify the codec effect. `roundtrip_wav` is the AAC bitstream of `mp4_aac64`, decoded and then re-wrapped as WAV. It carries the same codec output through a different container, with one additional int16 quantisation. It is not bit-identical to the direct MP4 decode, because the WAV path requantises the samples. Its median standardised feature difference is nevertheless $|\mathrm{SMD}| = 7.6 \times 10^{-5}$, and only 12 of the 436 descriptors exceed 0.05. That value is the noise floor of this study. Section VI-F uses this condition as an informationally empty perturbation rather than simply reporting that the difference is small.

Feature extraction over $7{,}442 \times 4 = 29{,}768$ files failed on exactly one clip in all four conditions. That clip is `1076_MTI_SAD_XX`, which the corpus documentation lists as containing no audio. The remaining 7,441 clips form the analysis set.

C. What the codec does to the descriptors

Fig. 2 characterises the coding manipulation before any model is involved. Two mechanisms can be distinguished, and they behave differently from each other.

MP3 abandons the top octave. Perceptual bit allocation quantises the highest band coarsely and sets masked bins to zero. As a result, `spectral_contrast_6_mean` moves from 16.38 to 46.54, and spectral flatness collapses at the same time, with its mean falling from 0.0189 to 0.0066, because the residual spectrum is far less noise-like than the original. Measured in training-set standard deviations, this shift is 38.6σ. The descriptor therefore arrives at inference time in a range that no model fitted on clean audio has ever seen.

AAC shifts frame alignment. Encoder priming lengthens every clip uniformly by 32.68 ms. This perturbs every frame-aligned temporal derivative a little, rather than destroying a single band. The largest AAC shifts are therefore all delta-MFCC means, led by `mfcc_d1_01_mean` at $\mathrm{SMD} = -4.16$. We raise one caveat here rather than in the limitations section, because it affects how these

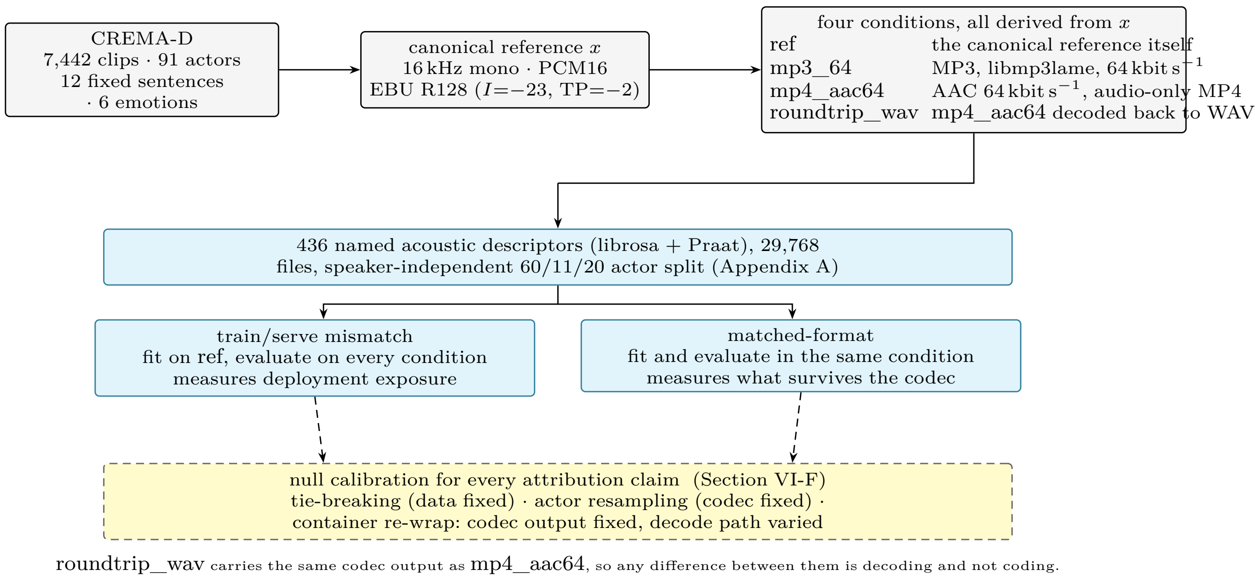

Fig. 1. The experimental design. Every condition descends from a single canonical reference x, so no comparison between formats can be confounded by a different resampling path or a different gain path. The same models are fitted twice. They are fitted once on uncompressed audio and then served compressed audio, and they are fitted a second time and served in the same condition they were trained on. The difference between those two readings is the first result of this paper. Every attribution claim is then evaluated against a floor measured on a manipulation that carries no coding information.

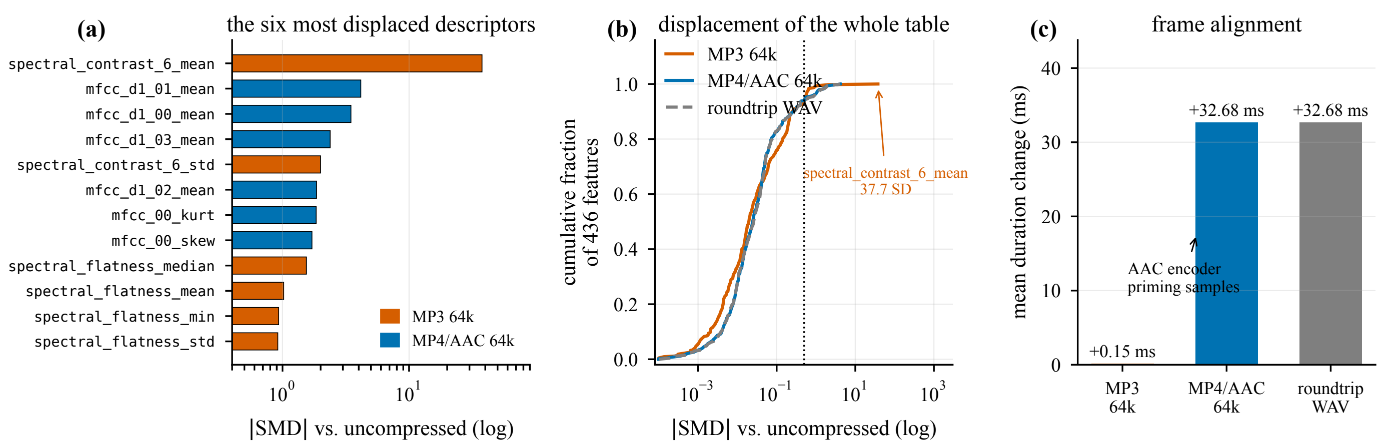

Fig. 2. The coding manipulation, measured on the feature table over 7,441 clips. (a) The six most displaced descriptors under each codec. The displacement caused by MP3 is concentrated on a single descriptor. spectral_contrast_6_mean, which summarises the 6 kHz to 8 kHz band, moves by 37.7 pooled standard deviations. The displacement caused by AAC is diffuse, and it lands on the delta-MFCC means, which are the frame-aligned temporal derivatives. (b) Empirical cumulative distribution function of the absolute standardised mean difference over all 436 descriptors. The median shift is negligible at roughly 0.02, but the tail is not. (c) AAC inserts encoder priming samples, which lengthen every clip by 32.68 ms. MP3 adds only 0.15 ms.

numbers should be read. Delta means sit near zero and have small standard deviations, so their standardised shifts are inflated relative to their absolute change. The duration measurement is the stronger evidence for the padding mechanism.

IV. Feature Representation

Each file is reduced to 436 named descriptors in four families. Every frame-level contour is collapsed to eight order statistics, which are the mean, standard deviation, minimum, maximum, median, interquartile range, skewness, and excess kurtosis. Appendix A defines every descriptor and enumerates every column. The families are summarised below.

- Spectral (114 descriptors). These are the centroid, bandwidth, roll-off at 85 % and 95 %, flatness, flux, entropy, slope, zero-crossing rate, RMS energy, seven-band spectral contrast, eight band-energy ratios, and explicit high-frequency survival ratios above 4 and 6 kHz.
- Cepstral (240 descriptors). These are 20 MFCCs together with their first and second temporal derivatives, Δ and $\Delta\Delta$.
- Prosodic (56 descriptors), computed in Praat through Parselmouth [23]. These are F_0 statistics and slope,

jitter in its local, RAP, PPQ5 and DDP variants, shimmer in its local, dB, APQ3, APQ5, APQ11 and DDA variants, harmonics-to-noise ratio, the formants F_1 to F_5 with their bandwidths, intensity, voiced-segment timing, and smoothed cepstral peak prominence (CPPS).
- Chroma (24 descriptors), together with two global descriptors, which are clip duration and peak amplitude.

Spectral and cepstral descriptors are computed with librosa [24] using a 512 point FFT, a 400 sample window, and a 160 sample hop. Chroma tuning is pinned to zero rather than estimated. An estimated tuning could itself shift under compression, and it would then become part of the effect we are trying to measure without our noticing.

We define one descriptor explicitly here, because it recurs in the results and because its formula is the same one that the trees of Section VI-D apply for a different purpose. Per-frame spectral entropy is the Shannon entropy of the power spectrum, normalised by its own bin count,

$$H_t = -\frac{1}{\ln K}\sum_{k=1}^{K} P_{k,t}\,\ln P_{k,t}, \qquad P_{k,t} = \frac{S_{k,t}}{\sum_{j=1}^{K} S_{j,t}}, \tag{1}$$

where $S_{k,t}$ is the power in bin k of frame t, and K is the number of bins. The normalisation constrains H_t to the interval $[0, 1]$ and makes it scale-free. The value approaches 1 for a frame whose energy is spread evenly across frequency, and it approaches 0 for a pure tone. Spectral entropy falls when an encoder sets masked bins to zero, and that is why it moves under MP3 before the MFCCs do.

A. Splitting

Actors are partitioned into training, validation and test sets in a 60/11/20 ratio, with sex balance preserved. This yields 4,905, 896 and 1,640 clips respectively. Speaker-independent splitting is not a mere refinement in this corpus. A variance decomposition over the feature table gives a mean η^2 of 0.178 for speaker against 0.098 for emotion, and 340 of the 436 descriptors are speaker-dominant. A random split over rows would therefore leak voice identity into the test set and inflate every score reported below. Item alignment is also enforced, meaning that the same clips appear in every condition, so that any comparison across conditions differs only in the coding.

B. Separability

397 of the 436 descriptors separate emotion at Bonferroni-corrected significance. The strongest single descriptors are mfcc_d1_00_std with $F = 1194$, mfcc_00_std with $F = 1159$, and prosody_intensity_std with $F = 1078$. All three are measures of variability rather than of mean level. Emotion in acted speech lives in the dynamics of the signal, and this fact recurs in every attribution result reported below. Ranking the families by mutual information inverts the ranking by column count, giving chroma at 0.098, then prosody at 0.087, then spectral at 0.082, and finally MFCC at 0.061. MFCCs therefore lead on column count and not on information per column. Fifty-one principal components are needed to account for 95 % of the variance, so this feature space is genuinely high-dimensional rather than a handful of latent factors carrying 436 different names.

V. Models and the Splitting Criterion

A. The zoo

We fit fourteen scikit-learn classifiers, which span tree ensembles [25], [26], kernel machines [27], linear and discriminant models, and instance-based and probabilistic baselines [28]. To these we add a PyTorch MLP with layer widths of 512, 256 and 128, batch normalisation, and dropout of 0.3. It is trained with AdamW under a cosine schedule, using class-weighted cross-entropy with label smoothing, and early stopping on validation balanced accuracy. This gives fifteen trained models in total.

B. Entropy and information gain

Trees are grown with the Shannon-entropy criterion rather than with the Gini criterion, so that the split rule reported in Section VI-D is exactly the rule these equations describe. For a node S carrying class weights w_c for each of the six emotions, with $w = \sum_c w_c$, the node entropy in bits is

$$H(S) = -\sum_{c=1}^{6} p_c \log_2 p_c, \qquad p_c = \frac{w_c}{w}. \tag{2}$$

A candidate test on descriptor f at threshold τ partitions S into $S_L = \{x : x_f \le \tau\}$ and S_R, and is scored by the information gain

$$\mathrm{IG}(S, f, \tau) = H(S) - \frac{w_L}{w} H(S_L) - \frac{w_R}{w} H(S_R). \tag{3}$$

At every node the tree selects the pair (f, τ) that maximises Eq. 3. Class weights are set to balanced, so each w_c is the inverse class frequency rather than a raw count. CREMA-D is close to balanced already, with 1,271 clips for each of five emotions and 1,087 neutral clips, so all of the weights are close to unity.

C. A worked example

Because the weights equalise the classes exactly, the root entropy of every tree in this paper is

$$H(S_{\mathrm{root}}) = -\sum_{c=1}^{6} \tfrac{1}{6} \log_2 \tfrac{1}{6} = \log_2 6 = 2.5850 bits. \tag{4}$$

On uncompressed audio the test that maximises the gain is prosody_intensity_std $\le$ 8.2184. This test sends $w_L = 3098.67$ of the $w = 4905.00$ weighted samples to the left branch and $w_R = 1806.33$ to the right branch,

TABLE I
Models trained and tested on uncompressed audio, evaluated on 20 held-out speakers in the six-way task. Chance performance is 0.167 and the human voice-only ceiling is 0.455.

Model	Acc.	Balanced acc.	Macro F_1	κ
ANN (PyTorch)	0.599	0.599	0.596	0.518
LightGBM	0.590	0.591	0.584	0.508
SVM (RBF)	0.585	0.586	0.581	0.502
LDA (shrinkage)	0.585	0.585	0.580	0.502
Linear SVM	0.583	0.584	0.577	0.500
XGBoost	0.581	0.582	0.575	0.497
MLP (scikit-learn)	0.573	0.574	0.569	0.488
HistGradientBoosting	0.573	0.573	0.566	0.487
Logistic regression	0.565	0.567	0.561	0.478
Random forest	0.532	0.535	0.517	0.439
AdaBoost	0.520	0.521	0.514	0.423
Extra trees	0.516	0.519	0.496	0.420
kNN ($k = 15$)	0.477	0.479	0.466	0.373
Gaussian NB	0.424	0.430	0.379	0.312
Decision tree	0.390	0.391	0.382	0.268
Dummy (majority)	0.171	0.167	0.049	0.000

with $H(S_L) = 2.3849$ bits and $H(S_R) = 2.0891$ bits. Substituting these values into Eq. 3 gives

$$\begin{aligned} \mathrm{IG} &= 2.5850 - \tfrac{3098.67}{4905.00}(2.3849) - \tfrac{1806.33}{4905.00}(2.0891) \\ &= 2.5850 - 1.5066 - 0.7693 \;=\; 0.3090 bits. \end{aligned} \tag{5}$$

The single most informative question that can be asked about a CREMA-D clip is therefore how much does its loudness vary, and the answer to that question resolves 12 % of the total label uncertainty. Every threshold, entropy and information gain reported below is computed in this way, and the complete node tables appear in Appendix B.

VI. Results

A. In-format performance

Table I gives the leaderboard on uncompressed audio. Two points matter for what follows, and neither of them concerns which model won.

First, a single decision tree reaches 0.391 balanced accuracy against 0.599 for the neural network. The decision tree is the one model in the zoo that a human can read from beginning to end, so intrinsic interpretability costs a third of the performance that is achievable on this task. That result argues against Rudin's prescription [16] in this specific setting, and it argues in favour of applying post-hoc attribution to a strong model. Section VI-G2 then shows what that trade buys and what it costs.

Second, the spread between the top eight models is less than three points, so the ordering of the leaderboard carries little information. The differences that matter in this paper range from 10 to 38 points, and they appear only when the coding condition changes.

B. Train/serve mismatch

Table II and Fig. 3 report what happens when a model fitted on clean audio is served coded audio. The result is categorical rather than graded. Every tree-based model stays within 0.014 balanced accuracy of its clean score on MP3, in either direction. Every model that consumes descriptor values as continuous magnitudes loses between 12 and 38 points.

The mechanism is readable because the descriptors are named. A tree asks whether spectral_contrast_6_mean exceeds a threshold near 17, and the answer to that question is the same whether the descriptor reads 16.4 or 46.5. A linear or kernel model instead multiplies that 38σ excursion directly into its decision function.

We name the two groups after the property that separates them rather than after a model family, because two members do not fit the obvious label. kNN computes distances and Gaussian NB evaluates per-descriptor likelihoods, and neither one multiplies a descriptor by a learned weight. What all eight models share is that the magnitude of a descriptor enters the decision continuously. The seven tree-based models only ever ask on which side of a threshold that magnitude falls.

MP4/AAC behaves differently. It costs every model something and no model very much, with losses of 0.7 to 11 points regardless of family. This pattern is consistent with the diffuse padding mechanism shown in Fig. 2(c). Container routing has no material effect. Across the fifteen models, the median difference between roundtrip and mp4_aac64 is 0.003 balanced accuracy and the maximum is 0.010, whereas the codec effects are one to two orders of magnitude larger.

C. Matched-format training

The mismatch result is a statement about deployment and not about the codec. To separate the two, every model was refitted within each condition, using the same actors, the same items, and the same hyper-parameters, with coded audio in both training and test. Table III and Fig. 4 give the outcome, and it is unambiguous.

The RBF-SVM scores 0.589 on MP3 when it is trained on MP3, against 0.586 on clean audio. That is a recovery of 38.6 balanced-accuracy points over its mismatched score of 0.203. Logistic regression recovers 33.0 points, LDA recovers 24.8 points, and the shallow MLP recovers 18.2 points. No model has a spread greater than 0.039 across the four conditions. The two largest spreads both belong to the single decision tree, whose variance is high for reasons that Section VI-F makes precise.

Three consequences follow from this table.

The codec does not remove the affective information. If MP3 at 64 kbit s^{-1} destroyed the acoustic correlates of emotion, then a model fitted on MP3 could not recover them. Such a model does recover them, to within three thousandths of the clean-audio score. The column of Table II that shows catastrophic losses therefore measures an artefact of deployment practice rather than a property of the codec.

The advantage held by the tree family exists only under mismatch. Under matched training, LightGBM leads the RBF-SVM by 0.008 on clean audio and trails it by 0.006 on

TABLE II
All models trained on uncompressed audio and served compressed audio. Balanced accuracy is measured on 20 held-out speakers. Models are grouped by how they consume a descriptor value. roundtrip is the AAC output re-wrapped as WAV, and it acts as the container control. Across the fifteen models its difference from mp4_aac64 has a median of 0.003 and a maximum of 0.010, reached by kNN.

	Model	ref	mp3_64	Δ MP3	mp4_aac64	Δ AAC	roundtrip
axis-aligned partitioning	LightGBM	0.591	0.586	−0.005	0.559	−0.032	0.562
	XGBoost	0.582	0.586	+0.004	0.560	−0.022	0.560
	HistGradientBoosting	0.573	0.559	−0.014	0.530	−0.043	0.533
	Random forest	0.535	0.532	−0.003	0.522	−0.013	0.522
	AdaBoost	0.521	0.511	−0.010	0.487	−0.034	0.484
	Extra trees	0.519	0.521	+0.002	0.512	−0.007	0.507
	Decision tree	0.391	0.382	−0.009	0.367	−0.023	0.371
magnitude-sensitive	kNN	0.479	0.362	−0.118	0.467	−0.012	0.457
	Gaussian NB	0.430	0.277	−0.152	0.437	+0.008	0.433
	Linear SVM	0.584	0.405	−0.179	0.471	−0.113	0.462
	MLP (scikit-learn)	0.574	0.383	−0.191	0.491	−0.083	0.494
	ANN (PyTorch)	0.599	0.354	**−0.246**	0.513	−0.087	0.512
	LDA	0.585	0.338	−0.247	0.507	−0.079	0.502
	Logistic regression	0.567	0.233	−0.334	0.472	−0.095	0.470
	SVM (RBF)	0.586	0.203	**−0.383**	0.499	−0.087	0.497

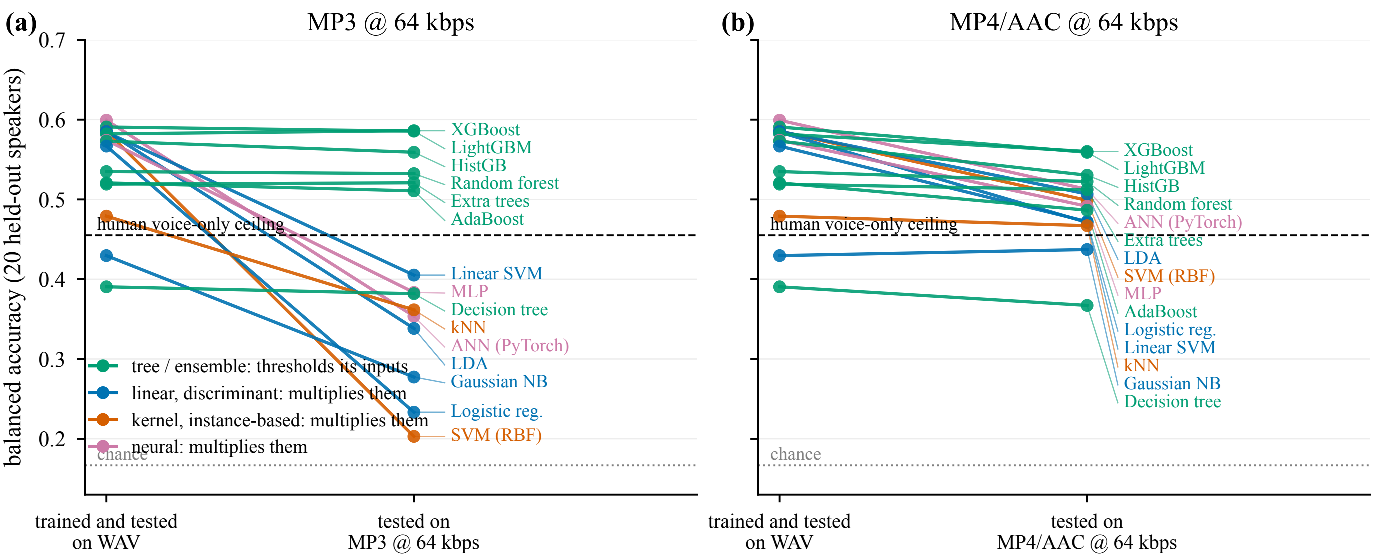

Fig. 3. Robustness to lossy coding under train/serve mismatch is determined by how a model consumes its inputs, and not by how accurate that model is. (a) Under MP3 at 64 kbit s^{-1} the tree family stays flat, while the magnitude-sensitive family falls through the human ceiling and towards chance. (b) Under MP4/AAC the same models degrade broadly and mildly, which is consistent with a diffuse frame-alignment perturbation rather than with the failure of a single band. All models were trained on uncompressed audio only.

TABLE III
Matched-format training. Balanced accuracy when a model is fitted and evaluated in the same condition. The rightmost column gives the spread across all four conditions. These values should be contrasted with Table II, where the same models lose up to 38 points.

Model	ref	mp3_64	mp4_aac64	roundtrip	range
LightGBM	0.591	0.583	0.582	0.585	0.008
SVM (RBF)	0.586	0.589	0.572	0.572	0.017
LDA (shrinkage)	0.585	0.586	0.579	0.584	0.007
MLP (scikit-learn)	0.574	0.566	0.569	0.558	0.017
Logistic regression	0.567	0.563	0.570	0.567	0.008
Random forest	0.526	0.519	0.524	0.523	0.007
Decision tree ($d = 12$)	0.408	0.379	0.398	0.369	0.039
Decision tree ($d = 3$)	0.407	0.408	0.389	0.389	0.019

MP3. The robustness ordering established in Section VI-B is therefore entirely a statement about inputs that fall outside the fitted range.

The fix is specific to one format rather than general. When the RBF-SVM is trained on mp4_aac64 and served MP3, it falls to 0.167, which is exactly chance. Matched-format training immunises a model against the codec it was trained on and against no other codec. That is why the descriptor-level repair described in Section VI-I is the more practical remedy when the delivery format is not known in advance.

D. What the tree splits on, per condition

Equal accuracy does not imply equal evidence. Because the models are fitted to named descriptors, we can ask the question directly. Given audio that supports the same

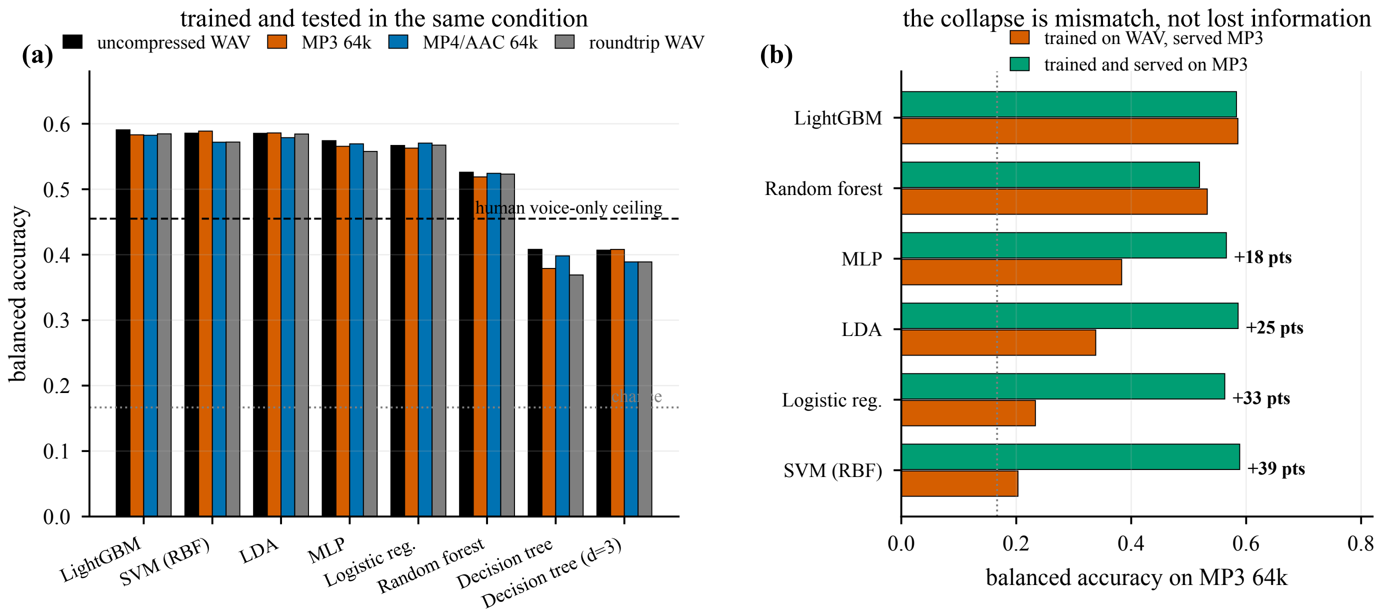

Fig. 4. Matched-format training removes the damage that mismatch creates. (a) Every model that is fitted and evaluated in the same condition stays within 0.008 to 0.039 balanced accuracy of its uncompressed score. This includes the two models that lose 33 and 38 points under mismatch. (b) The same comparison restricted to MP3. The affective information survives coding at 64 kbit s^{-1}. What fails is the mismatch between the distribution a model was fitted on and the distribution it is served.

TABLE IV
Root and depth-1 splits by condition, taken from the depth-3 entropy tree. H and IG are reported in bits, following Eq. 2 and Eq. 3. L denotes the branch taken when the root test is satisfied.

Condition	Descriptor	Threshold	IG
root $H = 2.5850 bits = \log_2 6$			
ref	prosody_intensity_std	8.2184	0.3090
mp3_64	prosody_intensity_std	8.2210	0.3090
mp4_aac64	prosody_intensity_std	8.3241	0.3064
roundtrip_wav	prosody_intensity_std	8.3241	0.3064
depth 1, branch L			
ref	mfcc_d1_00_std	3.9965	0.1415
mp3_64	mfcc_d1_00_std	4.0175	0.1398
mp4_aac64	mfcc_00_iqr	60.5853	0.1408
roundtrip_wav	mfcc_00_iqr	60.5802	0.1408

accuracy, does the model reach that accuracy by asking the same questions?

Fig. 5 and Table IV show the top two levels of an entropy tree fitted independently on each condition. Section VI-E carries the same comparison down to depth 6, and Appendix B tabulates every internal node of all four trees to depth 4.

We draw four observations from this table, in increasing order of interest.

The primary evidence does not change with the codec. All four trees select prosody_intensity_std at the root, and they extract almost exactly the same amount of information from it. The gain is 0.3090 bits under WAV and MP3, and 0.3064 bits under AAC. Whatever coding does to this feature space, it does not change the single most discriminative question that can be asked about a clip.

The threshold moves, and it moves in step with the mechanism. MP3 shifts the root threshold by 0.03 %, from 8.2184 to 8.2210. AAC shifts it by 1.29 %, to 8.3241. The standard deviation of intensity is computed over frames, and the AAC priming adds 32.68 ms of near-silent lead-in to every clip, which raises the dispersion of the intensity contour. The tree compensates by raising its threshold. This is the padding mechanism of Fig. 2(c) appearing as a number inside a decision rule.

The trees diverge at depth 1, under AAC and not under MP3. The uncompressed tree and the MP3 tree both ask about mfcc_d1_00_std, which is the standard deviation of the first temporal derivative of MFCC-0. The AAC and roundtrip trees abandon that descriptor in favour of mfcc_00_iqr, which is a static dispersion measure of the same coefficient. This is precisely what a frame-alignment perturbation should do. It corrupts the derivatives while leaving the distribution of the underlying contour intact, so the tree switches from a temporal statistic to a positional one at almost identical information gain, namely 0.1408 bits against 0.1415 bits. The model finds an equally good question that the codec has not damaged.

The container control reproduces at the level of tree structure. mp4_aac64 and roundtrip_wav carry the same codec output through different containers. Their trees select identical descriptors at every node of the top two levels. The root thresholds are 8.3241 in both cases, and the depth-1 thresholds are 60.5853 and 60.5802. Two conditions that differ only in container therefore produce the same model over these levels, and two conditions that differ in codec do not. Section VI-E shows how far that agreement extends, which is three levels rather than the whole tree. The distinction between container and codec

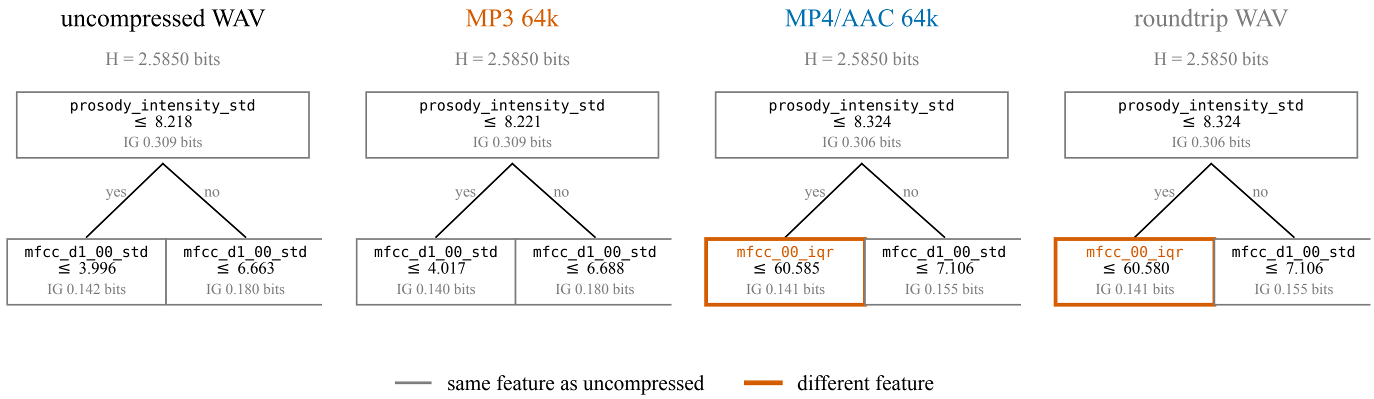

Fig. 5. The top two levels of the depth-3 entropy tree, fitted separately on each condition using the same actors and the same items. The identity of the root test does not change with the codec, and its threshold and information gain change very little. The trees begin to diverge one level below the root, and they diverge under AAC rather than under MP3. The AAC and roundtrip trees abandon the frame-aligned delta descriptor mfcc_d1_00_std in favour of the static dispersion measure mfcc_00_iqr, which is what the 32.68 ms priming shift predicts. AAC and roundtrip carry the same codec output through two different containers, and they select identical descriptors with thresholds that agree to four decimal places.

TABLE V
Descriptor agreement at path-aligned nodes, by depth, on the full training set. Each cell gives the number of nodes selecting the same descriptor out of the nodes that are internal in both trees. The rows above the rule carry no coding information.

Comparison	$d0$	$d1$	$d2$	$d3$	$d4$	$d5$
Tie-breaking (data fixed)	1/1	2/2	4/4	8/8	16/16	29/30
Container re-wrap (empty)	1/1	2/2	4/4	8/8	11/16	12/23
MP3 @ 64k	1/1	2/2	4/4	4/8	5/16	4/27
MP4/AAC @ 64k	1/1	1/2	2/4	0/8	1/16	0/24
Roundtrip WAV	1/1	1/2	2/4	0/8	0/16	0/23

was previously visible only as a difference in accuracy. Here it is visible as a difference in the learned decision rule itself.

E. How far down the tree the agreement survives

The comparison above stops at depth 1, which leaves the obvious question open. Agreeing on the root is agreeing on one node out of many, so the trees may be sharing only their first question and reorganising completely once the first-order evidence is spent. To answer that, the same tree was grown to depth 6 on each condition, which is 55 to 61 internal nodes per condition before the 25-sample leaf minimum stops further growth.

Two conditions are compared by aligning their nodes on the path from the root, written as a string of L and R branches. The path is the only identifier that means the same thing in two trees fitted on different data, because node indices simply record the order in which the trees were built. Growing deeper does not disturb what was already reported. The tree-building algorithm is greedy, and the leaf minimum is unchanged, so the top three levels of these deeper trees are node for node the trees of Table IV. Appendix B now enumerates all 31 internal nodes down to depth 4 for each condition.

Table V and Fig. 6 give the answer, and it is no in a precise and quantified way. Agreement decays with depth for every comparison, and the rate at which it decays is ordered by codec.

The estimator is not the explanation. Refitting on identical audio under a different tie-breaking seed leaves every node to depth 4 untouched and displaces one node of 30 at depth 5. The reproducibility of the fitting procedure is therefore not what limits agreement anywhere in this table. We note that this floor has to be averaged over a panel of seeds rather than read from one reseed, because most seeds leave the tree completely untouched while some displace a single deep node. A single reseed would have reported this floor as either perfect or one node short, purely by luck.

An informationally empty perturbation costs three levels. The container re-wrap holds the codec output fixed and changes only the decode path. It reproduces the reference tree exactly to depth 3, then falls to 69 % agreement at depth 4 and 52 % at depth 5. Below depth 3 the structure of this tree is therefore not a property of the audio at all, and the same discipline that Section VI-F applies to the importance ranking is required here.

Both codecs fall below that floor, and they fall in the order the mechanisms predict. MP3 reproduces the reference tree to depth 2 and AAC only to depth 0. Under paired actor resampling the ordering is unchanged and every gap is significant. The tie-breaking floor reaches a mean depth of 5.10 ± 1.21 and the empty container floor 2.30 ± 0.57 ($p = 1.2 \times 10^{-4}$, $d_z = 2.06$). MP3 reaches 0.90 ± 0.91, which is below the empty floor ($p = 3.5 \times 10^{-4}$, $d_z = 1.34$), and AAC reaches -0.05 ± 0.51 ($p = 8.7 \times 10^{-5}$, $d_z = 2.89$). AAC displaces the structure more than MP3 does ($p = 1.1 \times 10^{-3}$, $d_z = 1.15$), which reproduces the ranking result of Section VI-F on a completely different statistic.

The container control is exact at this level. mp4_aac64 and roundtrip_wav return the identical divergence depth in all 20 draws, so the paired test between them has no variance to work with and returns no statistic at all.

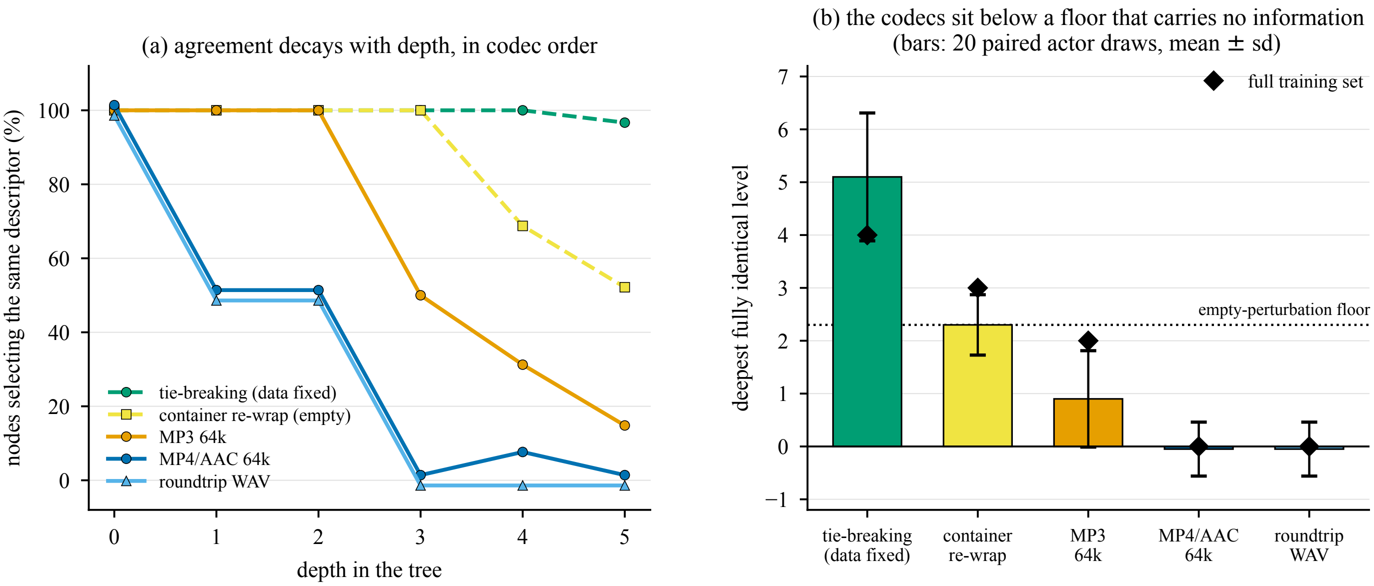

Fig. 6. Codec invariance of the tree structure does not survive past the top levels, and how far it survives is ordered by codec. (a) The percentage of path-aligned nodes at each depth that select the same descriptor in both trees. The AAC and roundtrip curves are identical at every depth, so they are drawn with a small vertical offset to keep both visible. (b) The deepest level to which two trees are node for node identical. Bars give the mean over 20 paired draws of 51 of the 60 training actors, with the floor and the effect measured inside the same draw, and diamonds give the value on the full training set. A value of 0 means the trees share only their root, and a negative value means even the root descriptor differs.

Two conditions carrying the same codec output through different containers are structurally indistinguishable from each other, while both differ from the reference at depth 1.

Two details are worth recording because they qualify how the agreement should be read. First, at depth 1 the AAC tree keeps `mfcc_d1_00_std` on the `R` branch and replaces it only on the `L` branch, which is why its depth-1 agreement is one node of two rather than none. The codec displaces the descriptor where the clip is quiet, and leaves it in place where the clip is loud. Second, the thresholds of the nodes that do still agree drift further apart with depth. The median relative threshold shift under MP3 grows from 0.03 % at the root to 15.3 % at depth 5, and the median absolute difference in information gain grows from zero at machine precision to 0.054 bits at depth 5. Agreement on a descriptor at depth 5 is therefore a much weaker statement than agreement on the same descriptor at the root.

The practical reading is that codec invariance in this feature space is a property of the first two or three questions a model asks, and not of the model as a whole. That is a narrower claim than Section VI-D on its own would support, and it is also the more useful one, because those first questions carry most of the information gain.

F. Null calibration of the importance ranking

The obvious next step is to compare the full importance rankings of the models across conditions, and the obvious result is a dramatic one. The tree's top-25 descriptors under MP3 share only 12 members with its top-25 descriptors under WAV, and the rank correlation over the descriptors that either tree uses is $\rho = 0.017$, which indicates no relationship at all. Read at face value, this says that coding has almost completely rewritten what the model considers important.

TABLE VI
Null calibration for the importance ranking of the decision tree. The rows above the rule are manipulations that carry no coding information, and the rows below it are the coding conditions. ρ is the Spearman rank correlation computed over the descriptors that at least one of the two fits actually used.

Comparison	top-25 kept	ρ	What it varies
Tie-breaking	23.5	+0.946	estimator only, data fixed
Actor bootstrap	8.0	−0.200	training actors, codec fixed
Container re-wrap	16.0	+0.184	decode path, codec output fixed
MP3 @ 64k	12.0	+0.017	codec
MP4/AAC @ 64k	9.0	−0.082	codec
Roundtrip WAV	10.0	−0.123	codec

That reading cannot be supported, and establishing why is the methodological contribution of this paper. The deep importance ranking of a decision tree is a fragile object. We therefore measure three floors, all of them on the same estimator that produced the effect.

Table VI and Fig. 7(a) show why the face-value reading fails. Resampling which actors the tree is fitted on, while holding the codec fixed at uncompressed, retains only 8.0 of the 25 descriptors. That is worse than any codec condition. When the comparison is made marginally, every codec effect sits inside the sampling floor, and no claim about the codec can be made at all.

The marginal comparison is the wrong one, because it varies two things at the same time. The floor is estimated on one training set and the effect is estimated on another.

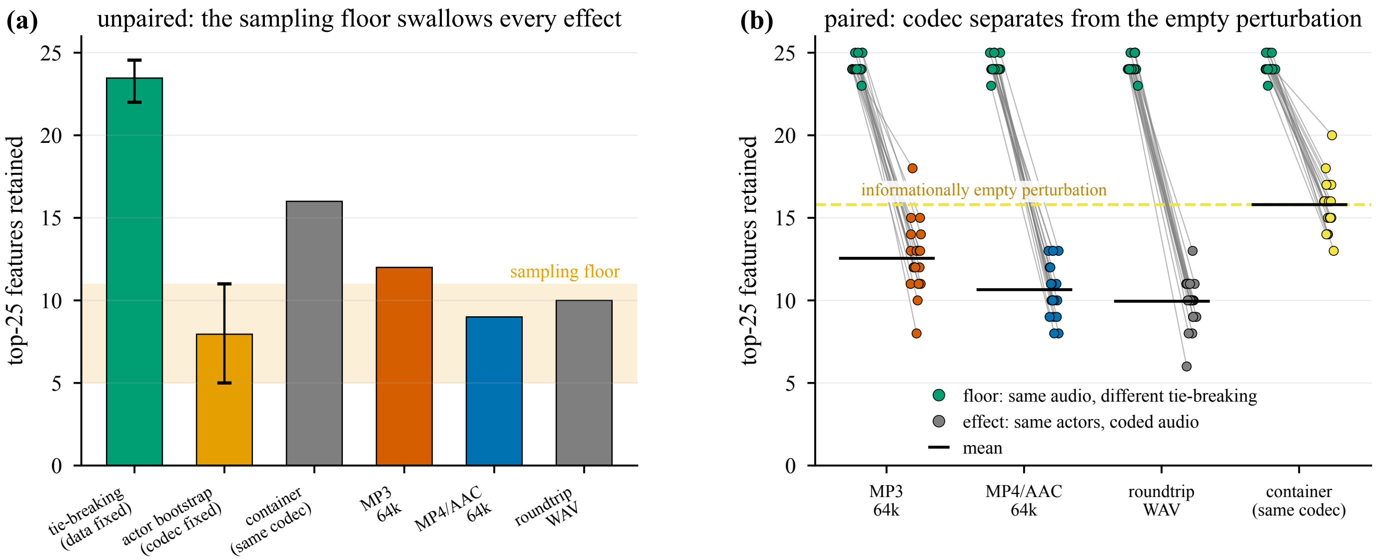

Fig. 7. A codec effect on a feature-importance ranking cannot be interpreted without a null. (a) When the comparison is made marginally, the actor-resampling floor is so wide that every codec effect falls inside it. (b) When floor and effect are measured inside the same resampled training set, using 20 draws of 51 of the 60 training actors and computing both quantities on the same draw, the codec effect separates cleanly. The decisive control is the rightmost group. An informationally empty container re-wrap already costs 8.4 of the 25 descriptors, so only the gap between that line and the codec bars can be attributed to the codec.

TABLE VII
Paired within-draw comparison over 20 draws of 51 of the 60 training actors. The floor is a refit on the uncompressed audio of the same actors under different tie-breaking. The p values come from a Wilcoxon signed-rank test on the paired per-draw difference.

	top-25 kept	ρ	p vs. floor
floor (same audio, refit)	24.15	+0.958	n/a
Container re-wrap (empty)	15.80	+0.196	8.1×10^{-5}
MP3 @ 64k	12.55	−0.068	8.4×10^{-5}
MP4/AAC @ 64k	10.65	−0.137	8.3×10^{-5}
Roundtrip WAV	9.95	−0.126	7.9×10^{-5}

We therefore measure both quantities inside the same draw. On each of 20 draws, 51 of the 60 training actors are sampled without replacement. The tree is then fitted twice on the uncompressed audio of those actors under different tie-breaking, which gives the floor for that draw, and once on the coded audio of the same actors, which gives the effect for that draw. Sampling variance is held fixed by construction, and no actor is duplicated.

Table VII and Fig. 7(b) give the result, and it has three layers.

The estimator is reproducible. With the data held fixed and only the tie-breaking varied, the tree retains 24.15 of its 25 top descriptors, at $\rho = 0.958$. Whatever else in this analysis is fragile, the fitting procedure is not.

An informationally empty perturbation is expensive. The container re-wrap changes no codec information whatsoever. It is the same AAC bitstream, decoded in the same way, and it differs only by a single int16 requantisation whose median effect on the features is 7.6×10^{-5} standardised units. It nonetheless costs 8.35 of the 25 top descriptors, moving the retention from 24.15 to 15.80 at $p = 8.1 \times 10^{-5}$. A ranking that turns over by a third under a manipulation carrying no information cannot be read as a description of what the model needs.

The codec effect is real, but it is roughly two fifths of the size it appears to be. Measured against the empty-perturbation floor of 15.80, MP3 retains 12.55 descriptors ($p = 1.1 \times 10^{-3}$, $d_z = 1.20$) and AAC retains 10.65 ($p = 1.2 \times 10^{-4}$, $d_z = 2.39$). Of the 13.5 descriptors that AAC appears to displace relative to a clean refit, 8.35 are therefore attributable to generic sensitivity to perturbation, and only 5.15 are attributable to the codec. A study that reported the raw figure would over-attribute the effect by a factor of about 2.6.

The direction of the codec-specific component agrees with Section VI-D. AAC displaces more of the ranking than MP3 does, with retentions of 10.65 against 12.55 ($p = 4.2 \times 10^{-3}$, $d_z = 0.84$, paired). That ordering is the reverse of the accuracy ordering under mismatch, where MP3 is catastrophic and AAC is mild. Coding that perturbs every temporal derivative slightly reorganises a tree more than coding that destroys one band which the tree can simply threshold around. Section VI-D shows the same thing structurally, because it is the AAC tree and not the MP3 tree that replaces its depth-1 descriptor.

G. Post-hoc attribution and its disagreements

Attribution was computed on the held-out speakers. We used TreeSHAP for the ensembles [11], LinearSHAP for the linear models, LIME for local surrogates [9], permutation importance, partial dependence, and a depth-4 global surrogate tree. For the neural network we used six methods from Captum [14], which are Saliency, Input×Gradient, Integrated Gradients [12], DeepLIFT [13], GradientSHAP and Feature Ablation.

TABLE VIII
Share of total attribution mass by descriptor family, measured on uncompressed audio. Column share is the proportion of the 436 descriptors that each family contributes. All rows use SHAP except the ANN row, which uses integrated gradients.

Model	MFCC	spectral	prosody	chroma	global
column share	0.550	0.261	0.128	0.055	0.005
XGBoost	0.566	0.207	0.178	0.033	0.016
LightGBM	0.550	0.209	0.189	0.030	0.021
Random forest	0.438	0.315	0.205	0.032	0.011
Decision tree	0.436	0.183	0.306	0.034	0.041
Logistic regression	0.471	0.306	0.156	0.062	0.005
ANN (int. grad.)	0.542	0.238	0.153	0.057	0.009

1) Convergent evidence across families: On uncompressed audio, five model families and four attribution algorithms select the same acoustic evidence. The top SHAP descriptors for XGBoost and LightGBM are identical in their first five entries, which are mfcc_d1_00_std, mfcc_d2_00_std, duration_s, prosody_intensity_std and mfcc_00_std. The decision tree leads with prosody_intensity_std, which matches its own root split. All six gradient methods applied to the network return mfcc_d1_00_std, mfcc_d2_00_std, prosody_cpps and duration_s. The two leading descriptors are the standard deviations of the first and second derivative of MFCC-0. They measure how variable the rate of change of overall loudness is, which is a direct formalisation of vocal agitation.

Attribution mass is spread across the families rather than concentrated in one of them, as Table VIII shows. The pattern matches the mutual-information result of Section IV. MFCCs hold 55 % of the columns and take roughly 55 % of the attribution, so they carry no advantage per descriptor. The single decision tree is the one outlier, because it leans more heavily on prosody. That reliance is what makes the tree readable, and it is also what limits its accuracy.

The per-emotion attributions are acoustically sensible. CPPS, which correlates with breathiness, drives both sadness and neutrality. The median F_0 drives fear, movement in F_2 drives disgust, and loudness dynamics drive anger. The network was given none of this information in advance.

2) The methods disagree with each other: Fig. 8(a) presents an uncomfortable result. For the simpler models, SHAP agrees closely with a model's intrinsic importance, reaching $\rho = 0.98$ for both the decision tree and logistic regression. LIME and permutation importance are close to unrelated to that intrinsic importance. On the decision tree, LIME and SHAP rank descriptors at $\rho = 0.008$, which is no relationship at all, and yet the two methods still share 10 of their top 20 descriptors. This reproduces the disagreement problem documented by Krishna et al. [7], and it reproduces it on a model simple enough that approximation error cannot be blamed.

Panel (b) sharpens this point rather than softening it. The six gradient-family methods applied to the network agree almost perfectly, at $\rho \geq 0.94$ for every pair, with Integrated Gradients and DeepLIFT reaching 0.998. High agreement within a single method family is therefore not evidence of correctness. These methods share a mathematical lineage, and their consensus reflects that shared lineage. The practical consequence is that a claim about which acoustic descriptors matter, made from one method alone, does not reproduce across methods. Only the descriptors that survive all four independent views should be reported as robust. In this study that set consists of mfcc_d1_00_std, prosody_intensity_std, duration_s, prosody_cpps and prosody_f0_median.

Two further calibrations belong with this result. First, a depth-4 global surrogate reproduces the predictions of its target model only 57 to 74 % of the time. The fidelity is 73.9 % for the random forest, 63.4 % for LightGBM, 62.1 % for XGBoost, 58.7 % for the SVM, and 57.1 % for logistic regression. Surrogate rules therefore summarise a majority of a model rather than the whole model, and reporting them without their fidelity would overstate what has been explained. Second, the attributions produced for the network pass the Adebayo model-randomisation check shown in Fig. 9. The correlation between attributions from the trained network and attributions from a randomly initialised network of identical architecture ranges from -0.011 to 0.206 across all six methods. These explanations therefore describe the trained model and not merely the input distribution.

3) An estimator that failed loudly: The RBF-SVM has no exact explainer, so we applied KernelSHAP to it with 100 sampled coalitions over 436 descriptors. KernelSHAP fits a weighted least-squares regression with one unknown per descriptor. With 100 coalitions available for 436 unknowns, that system is rank-deficient, and the result diverged. The maximum mean |SHAP| value reached 2.1×10^{12} against a median of 5.3×10^{-4}, and entire descriptor families were assigned exactly zero. There is a cheap way to detect this failure. A well-behaved additive attribution computed over 7,441 clips does not assign a whole family a value of exactly 0.000.

We exclude the KernelSHAP ranking for the SVM from every conclusion in this paper, treating it as an estimation artefact, and the implementation now raises an error when the coalition budget does not exceed the descriptor count. The SVM therefore contributes accuracy and robustness results but no attribution row. Its trustworthy views are permutation importance and LIME, whose top descriptors reproduce the consensus that every other model reaches.

H. Attribution stability under coding

Table IX reports the stability of SHAP rankings under train/serve mismatch. The attributions of the tree and linear models barely move, while those of the network move substantially.

These rows measure something different from what Section VI-F measures, and the distinction matters. Here the model is held fixed and only its input changes, so

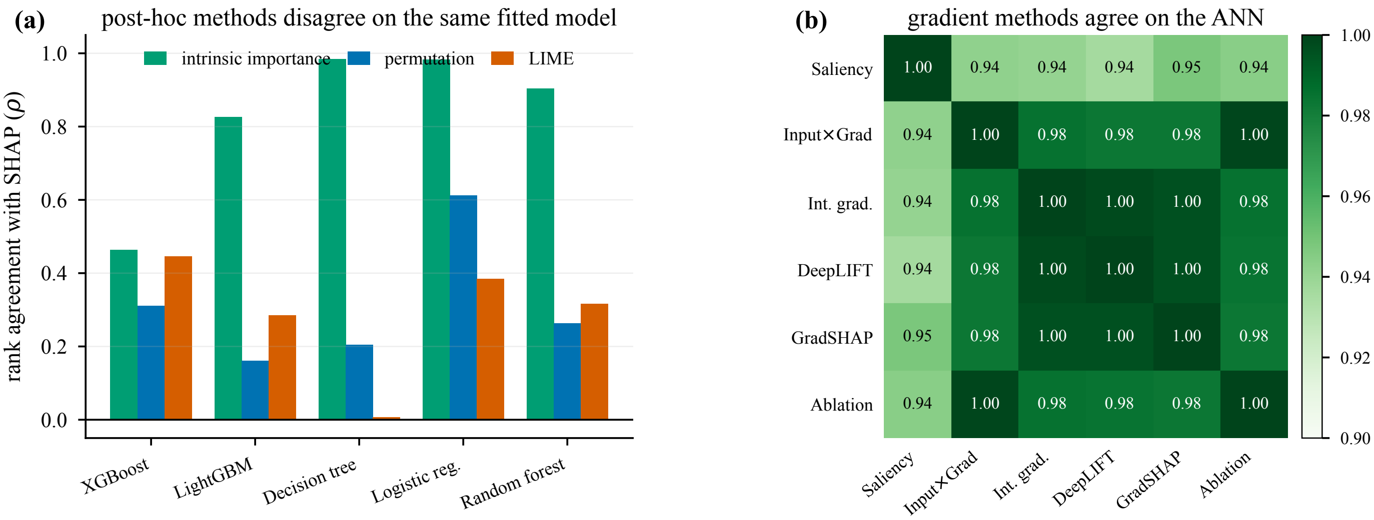

Fig. 8. Agreement between explanation methods, measured as Spearman rank correlation over all 436 descriptors. (a) Post-hoc methods applied to the same fitted tabular model. SHAP tracks intrinsic importance closely for the simpler models, while LIME and permutation importance are nearly unrelated to it. On the decision tree, LIME and SHAP agree at only $\rho = 0.008$. (b) The six methods of the gradient family, applied to the network, agree almost perfectly, at $\rho \geq 0.94$ for every pair. This shows that the disagreement in panel (a) is a property of the method families rather than of this feature space.

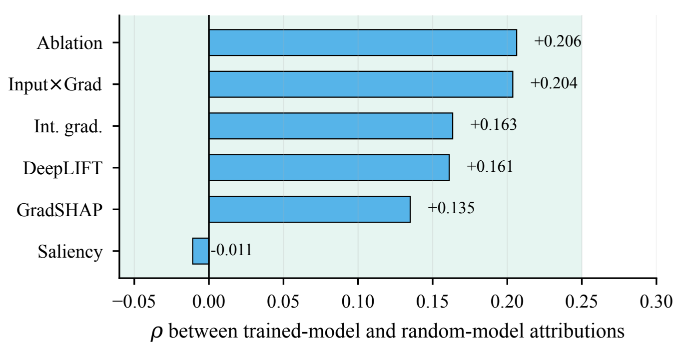

Fig. 9. Model-randomisation sanity check [6] for the six methods of the gradient family. A method whose attributions correlate with those of a randomly initialised network of identical architecture is describing the input distribution rather than the model. All six methods sit near zero, so this failure mode does not apply here.

TABLE IX
Attribution stability under mismatch. Values are Spearman ρ over all 436 descriptors, measured against the uncompressed condition, with the number of top-25 descriptors retained given in parentheses. All rows use SHAP except the ANN row, which uses integrated gradients.

Model	mp3_64	mp4_aac64	roundtrip	Δ acc. MP3
Decision tree	0.995 (24)	0.988 (21)	0.988 (21)	−0.009
XGBoost	0.991 (25)	0.984 (22)	0.984 (22)	+0.004
Random forest	0.990 (25)	0.976 (25)	0.975 (25)	−0.003
LightGBM	0.988 (23)	0.983 (22)	0.982 (22)	−0.005
Logistic regression	0.987 (22)	0.942 (21)	0.941 (21)	**−0.334**
ANN (int. grad.)	0.677 (11)	0.891 (16)	0.887 (16)	−0.246

the comparison is between two attributions of a single fitted object. In Section VI-F the model is refitted, so the comparison is between two separately learned objects. The first comparison is stable, because the split structure of a fitted tree does not move when it is fed slightly different numbers. The second is not stable, because the fitting procedure can arrive at an equally good structure by a different route.

Even so, the accuracy figure understates the attribution figure throughout the table. XGBoost loses 2.2 accuracy points on AAC and still swaps 3 of its top 25 descriptors. The network loses 8.7 points and swaps 9. A benchmark based on accuracy alone would call the first change negligible and the second change modest.

Which descriptors replace which is readable only because the descriptors are named. Under MP3, the descriptors entering the network's top 25 are almost all encoder artefacts. They are spectral_flatness in five different statistics, together with spectral_contrast_6_mean. The descriptors leaving the top 25 are genuine measures of voice quality and prosody, namely prosody_cpps, prosody_f0_skew, prosody_f2_mean, prosody_f2_std and prosody_f3_bw_median. Under mismatch, the model stops listening to the voice and starts listening to the encoder.

1) *A blind spot in global attribution metrics:* The logistic regression row of Table IX is the most important negative result in this paper. Its SHAP ranking is essentially unchanged under MP3, at $\rho = 0.987$ with 22 of the 25 top descriptors retained, while its balanced accuracy collapses from 0.567 to 0.233. The coefficients of the model did not change. Only the inputs moved out of range, and a global ranking based on mean |SHAP| is largely blind to that kind of movement.

This is a caution about the method rather than about the model. A global attribution-stability metric can report that nothing has changed while the classifier has effectively stopped working. Any deployment that monitors explanation stability as a proxy for model health must therefore pair that monitor with input-range monitoring or with local, per-instance attribution. The global ranking on its own will not raise the alarm.

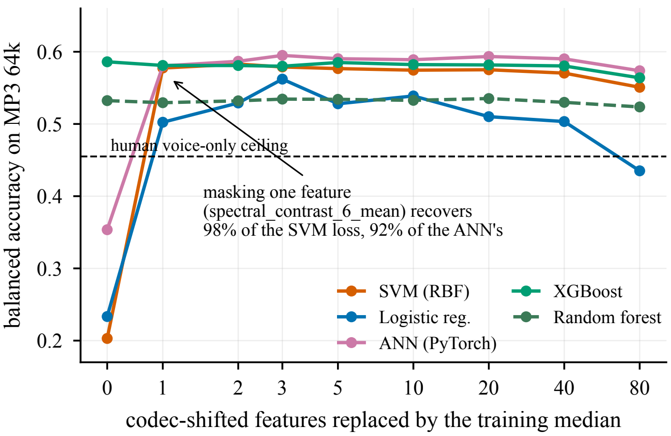

Fig. 10. Neutralisation on MP3 at 64 kbit s^{-1}, which replaces the most codec-shifted descriptors with their training medians. Masking a single descriptor recovers almost the entire collapse for every affected model. The tree models were never affected and stay flat throughout. Beyond roughly 20 descriptors the masking begins to destroy real evidence. On uncompressed audio the same masking costs nothing out to 20 descriptors, which confirms that these columns never carried load-bearing evidence.

TABLE X
Balanced accuracy on MP3 as the most codec-shifted descriptors are replaced by training medians. The italic row gives each model's own clean-audio score. The tree columns stay flat because those models had nothing to recover.

Masked	SVM	Log. reg.	ANN	MLP	XGBoost	Dec. tree
clean	0.586	0.567	0.599	0.574	0.582	0.391
0	0.203	0.233	0.354	0.383	0.586	0.382
1	0.578	0.502	0.580	0.552	0.581	0.382
2	0.583	0.529	0.587	0.554	0.581	0.391
3	0.579	0.562	0.595	0.565	0.580	0.391
10	0.575	0.539	0.589	0.562	0.582	0.394
80	0.551	0.435	0.574	0.523	0.564	0.396

I. Diagnosis and repair

Because the descriptors are named, the mismatch failure is not only observable but also addressable. We ranked the descriptors by how far each codec moves them in units of training standard deviation. We then replaced them progressively with training medians, in both the compressed test set and the clean test set. The clean column controls for whatever real signal the masking itself destroys.

The single masked descriptor is spectral_contrast_6_mean. Masking that descriptor alone recovers 98 % of the 38.3-point loss suffered by the RBF-SVM and 92 % of the 24.6-point loss suffered by the network. Masking three descriptors brings logistic regression to 0.562 against its clean-audio score of 0.567.

Two controls make this a diagnosis rather than a coincidence. First, on uncompressed audio the identical masking costs nothing out to 20 descriptors. The clean control for the network sits between 0.607 and 0.609, which is marginally above its unmasked baseline of 0.599. These columns therefore never carried load-bearing evidence. They were the channel through which the encoder's fingerprint entered the model. Second, the AAC case behaves as its different mechanism predicts, in that no single descriptor dominates. The best recovery for the network arrives at 10 descriptors masked, moving it from 0.513 to 0.560, and the best recovery for the SVM arrives at 40 descriptors masked, moving it from 0.499 to 0.542.

The diagnosis is therefore covariate shift on a small set of identifiable descriptors rather than a loss of emotional information. That is exactly what Section VI-C establishes independently, by a completely different route.

VII. Discussion

A. Three questions that are usually one

This study separates three questions that a single accuracy number runs together. The first is whether the codec destroys the information. It does not, because matched-format training recovers 38.6 points and lands within 0.003 of the clean score. The second is whether the codec breaks deployed models. It breaks them severely, provided that those models consume descriptor magnitudes and were fitted on clean audio. The third is whether the codec changes what the model learns. It does, and the change spares the leading questions. The root test survives both codecs intact, MP3 leaves two further levels untouched, and everything below that is reorganised. Even the reorganisation is smaller than the raw numbers suggest, and only a properly nulled comparison can say by how much.

An evaluation that reports only the second question, which is what cross-format benchmarking does, would conclude that MP3 at 64 kbit s^{-1} is unusable for SER. The first question shows that this conclusion is really about the pipeline and not about the codec.

B. Null calibration is not optional

The container re-wrap changes no coding information at all, and it still turns over a third of the tree's top-25 ranking. It also destroys the node-for-node identity of the tree below depth 3. Any claim of the form "manipulation M changed which features the model uses" is therefore impossible to interpret without a perturbation that is matched in every respect except informational content. We propose this as standard practice for feature-attribution claims in affective computing. The cost is one additional condition and one additional model fit. The benefit is that the claim becomes falsifiable. In our case the benefit is also that the claim shrinks by a factor of 2.6 rather than being reported at face value.

The requirement is not specific to one statistic. We applied the same empty control to an importance ranking in Section VI-F and to the tree's split structure in Section VI-E, and in both cases it absorbed a large part of what looked like a codec effect. The two statistics are computed from different objects and they agree on the ordering of the two codecs, which is the strongest evidence we have that the codec-specific remainder is real.

The paired design matters as much as the null itself does. When the comparison was made marginally, the actor-resampling floor was wide enough to swallow every codec effect, as Fig. 7(a) shows, and we would have reported nothing at all. Measuring the floor and the effect inside the same resampled training set removes sampling variance by construction and recovers a clean answer.

C. What a threshold buys, and what it does not

Under mismatch, axis-aligned models are almost perfectly robust to a 38σ excursion, because a threshold test does not care how far past the threshold a value lands. That is a real engineering advantage and a cheap one. However, Section VI-C shows that the advantage disappears under matched training, and Sections VI-E and VI-F show that the learned structure of the same models is more sensitive to a small diffuse perturbation than to a large concentrated one. Under matched training the depth-6 tree loses 2.2 balanced-accuracy points on MP3 and 3.8 on AAC, so the two codecs are 1.6 points apart. That small accuracy gap accompanies a two-level gap in the depth to which each reproduces the reference structure. A threshold protects the decision, and it does not protect the evidence.

D. Auditing and the archive problem

Per-instance rationales are increasingly presented to operators and auditors. Our results indicate that such rationales are not invariant across a transcoding step, even when that step leaves the decision invariant. Consider an audit conducted on archived compressed material while inference runs on uncompressed input, or the same situation reversed. In either case the decisions will agree while the explanations will not. An explanation is not reproducible unless the delivery format forms part of the record, and to our knowledge no SER benchmark currently reports it.

Section VI-E sharpens what an operator can safely be told. A rationale that cites the root test is reproducible across every condition we tested. A rationale that cites a rule several levels down is not, and neither is one that reports a ranked list of secondary features. How far past the root a rationale may go is itself codec-dependent, since MP3 supports two further levels and AAC supports none. Where a rationale has to be defensible, it should be truncated at the depth to which the structure has been shown to reproduce for the format actually in use.

E. What named descriptors buy

The repair described in Section VI-I has no counterpart in a system whose front end is learned. Identifying spectral_contrast_6_mean as the contaminated channel required attribution over units that carry acoustic meaning, and neutralising it required reaching into the representation itself. Reading the codec's frame-alignment mechanism off a decision threshold in Section VI-D required the same thing.

This is a narrower claim than Rudin's [16]. We are not arguing that interpretable models should replace strong ones. Table I shows the single interpretable model in our zoo costing a third of the achievable performance, and the best model overall is a neural network. We are arguing that a named-descriptor front end retains a capability for diagnosis and repair that end-to-end pipelines lack, and that in deployments with heterogeneous codecs this capability has measurable value. In our case that value was 98 % of a 38-point loss, recovered by masking a single column.

VIII. Limitations

1) The emotion is acted rather than spontaneous. CREMA-D actors perform prescribed emotions on 12 fixed sentences. The absolute accuracies reported here do not transfer to spontaneous speech, and the human voice-only ceiling of 45.5 % shows that the labels themselves are only loosely recoverable from audio.
2) The headline tables use one split rather than repeated cross-validation. Table I through Table III come from a single deterministic 60/11/20 partition over actors. Differences of one or two points should not be over-read. The null-calibration analysis of Section VI-F does resample, using 20 draws, and it is the only place where we make a claim that depends on resampling.
3) We test two codecs at one bitrate. Only MP3 and AAC at $64\,\mathrm{kbit\,s^{-1}}$ were tested. The MP3 mismatch result is severe partly because $64\,\mathrm{kbit\,s^{-1}}$ is an aggressive setting for that codec. Higher bitrates would probably show a gradient. Opus, which is now the dominant conversational codec, remains untested here.
4) Summary statistics discard time. Every frame contour is collapsed to eight order statistics, so no attribution reported here can localise when in a clip the evidence sits. The AAC padding mechanism is inferred from duration and from which descriptors move, and it is not observed directly in time.
5) The container control is not a perfect null. roundtrip_wav differs from mp4_aac64 by an int16 requantisation, so it is an almost empty perturbation rather than an exactly empty one. The error runs in the conservative direction, because a truly empty perturbation would have a lower floor and would make the codec effects look larger. A bit-exact container null would nevertheless be a stronger instrument.
6) The null calibration covers a single estimator. Section VI-F calibrates the decision tree, which we chose because it is the readable model of this paper and the one whose ranking is most obviously fragile. We do not establish whether ensemble or gradient attributions require floors of the same size, although the fixed-model comparison in Table IX suggests that they are more stable.
7) The depth profile is one tree, not an ensemble. The comparison of Section VI-E tracks a single tree with

a 25-sample leaf minimum, so the deeper nodes it compares are fitted on a few hundred clips each and would not survive in an ensemble that averages over them. The profile describes how far a readable structure is reproducible, and it is not a claim about how far a codec perturbs a boosted model.

8) The attribution comparisons are global. Spearman ρ and top-k overlap computed over the mean |attribution| are coarse instruments, and Section VI-H1 shows that they can miss a total collapse in accuracy. A local, per-instance stability analysis would be stronger, and we do not attempt one.
9) Speaker dominance limits what an importance ranking can mean. With a mean η^2 of 0.178 for speaker against 0.098 for emotion, a substantial part of any importance ranking describes which voices happen to be in the training set. The actor-bootstrap floor in Table VI quantifies this contribution, and it is large.
10) The work is exploratory rather than confirmatory. The per-condition and null-calibration analyses were designed after we had inspected the cross-format results. Confirmatory status would require preregistered replication.

IX. Future Work

Three directions follow from specific results reported above. First, a bit-exact container null would tighten the floor of Section VI-F, and it is cheap to obtain by comparing two lossless containers that carry identical PCM. Second, extending the null calibration to SHAP and integrated-gradients rankings under refitting would establish whether the fragility we document is specific to trees or general to feature attribution. Third, a bitrate ladder that includes Opus alongside MP3 and AAC would turn the two-point comparison of Section VI-D into a dose-response curve for structural displacement, with the depth profile of Section VI-E as its natural readout: the depth to which a structure reproduces is a single number per condition and is cheap to compute for a ladder of bitrates.

Beyond emotion, the tasks of speaker identification, vocal stress detection and voice-quality assessment rely on precisely the spectral detail that perceptual coding discards. The matched-format result may not transfer to them. A task whose evidence lives in the top octave has less to fall back on than a task whose evidence lives in loudness dynamics.

X. Conclusion

We asked whether perceptual audio coding changes what a speech emotion model relies on. We answered that question with models whose evidence is a named acoustic descriptor rather than a latent dimension.

Coding does not remove the affective information. A kernel machine that scores 0.203 on MP3 when it is fitted to clean audio scores 0.589 when it is fitted to MP3 itself, against 0.586 on clean audio. The catastrophic cross-format losses reported in the literature, and those in our own Table II, are caused by train/serve mismatch. They can be repaired either by matched training or by neutralising a single contaminated descriptor.

What coding does change is the structure that the model learns. An entropy tree keeps the same root question in every condition and extracts the same information from it. AAC nevertheless moves the root threshold by 1.3 % and replaces the depth-1 descriptor with one that its own frame-alignment perturbation leaves intact. Extending the comparison to depth 6 shows that this invariance is a property of the first questions the model asks and not of the whole tree. MP3 reproduces the reference structure to depth 2 and AAC only to depth 0, whereas the two conditions carrying identical codec output through different containers reproduce each other to depth 3 and return the same divergence depth in all 20 resampled draws.

The size of that change is easy to overstate, and the discipline that prevents the overstatement is the most transportable result in this paper. An informationally empty container re-wrap already displaces 8.35 of a tree's 25 most important descriptors. Only the gap beyond that floor can be attributed to the codec, and that gap is 3.25 descriptors for MP3 and 5.15 for AAC. We would ask reviewers of affective computing systems to require, alongside any claim that an intervention changed a model's explanation, a measurement of the divergence induced by an intervention that carries no information. Without such a measurement the claim cannot be evaluated. With it, our own claim shrinks by a factor of two and a half and becomes defensible.

Data Availability Statement

CREMA-D is publicly available from its authors under the terms stated in [22]. All derived artefacts are released with this paper. These comprise the stimulus construction scripts and condition manifests, the JSON schema of the 436-descriptor table over 29,768 rendered files, the per-condition and null-calibration result files, and the JSON and CSV files from which every table and figure is generated. The scripts that generate the figures and the appendices read only these committed artefacts, and they hard-code no numbers. The feature table itself occupies 243 MB, and both it and the rendered audio can be regenerated from the committed extraction code and configuration.

Ethics Declaration

This study involves no human subjects and no personally identifying data. It uses a published corpus of consenting professional actors performing scripted sentences, which was collected and released for research use by its original authors under their own ethical approval [22]. No new recordings were made. We note that speech emotion recognition is a dual-use technology. The findings reported here concern robustness and the validity of explanations, and if anything they are cautionary about deploying such systems on heterogeneous audio without format-aware validation. Nothing in this paper should be read as evidence that automatic emotion inference is accurate, fair, or appropriate in consequential settings.

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Appendix A
Complete Feature Inventory

Every stimulus is reduced to 436 named columns. Frame-level contours are summarised by eight order statistics (mean, standard deviation, minimum, maximum, median, interquartile range, skewness, excess kurtosis). Descriptors marked with a shorter list are summarised only by those statistics, and scalars are already clip-level. Naming is <family>_<descriptor>_<statistic> throughout, which is what makes an attribution over a column a claim about an acoustic quantity.

Analysis frames are 512 point FFT, 400 sample window, 160 sample hop at 16 kHz, which gives 25 ms windows every 10 ms. Praat measures use a 75 Hz to 500 Hz pitch range.

A. Descriptors and their definitions

B. Enumerated column names

The complete inventory, in the order the extractor emits it.

Appendix B
Per-Condition Tree Splits

The tables below list every internal node down to depth 4 of the entropy tree of Section VI-D. The tree was fitted independently on each condition, using the same actors, the same items and the same hyper-parameters. The Path column reads from the root, and L denotes the branch taken when the parent's test is satisfied. Entropies are given in bits. The value n is the raw sample count, and the information gain is computed on the class-weighted counts (Eq. 3).

A path missing from a table is one the tree stopped splitting, because a child would have fallen below the 25-sample leaf minimum. Comparing the same path across

TABLE XI
Spectral descriptors. 114 columns of 436.

Descriptor	Definition	Statistics	n
Spectral centroid	First moment of the power spectrum. It is the frequency about which spectral energy is balanced. Correlates with perceived brightness.	mean, std, min, max, median, IQR, skew, kurtosis	8
Spectral bandwidth	Power-weighted second moment about the centroid. It measures how widely energy is spread in frequency.	mean, std, min, max, median, IQR, skew, kurtosis	8
Spectral roll-off (85 %)	Frequency below which 85 % of the frame's spectral energy lies.	mean, std, min, max, median, IQR, skew, kurtosis	8
Spectral roll-off (95 %)	As above at 95 %. Directly sensitive to codec low-pass behaviour.	mean, std, min, max, median, IQR, skew, kurtosis	8
Spectral flatness	Geometric mean of the magnitude spectrum divided by its arithmetic mean. It approaches 1 for noise-like frames and 0 for tonal ones, and it collapses when an encoder sets masked bins to zero.	mean, std, min, max, median, IQR, skew, kurtosis	8
Zero-crossing rate	Fraction of adjacent sample pairs with a sign change. It is a cheap proxy for voicing and frication.	mean, std, min, max, median, IQR, skew, kurtosis	8
Frame RMS energy	Root-mean-square magnitude per frame, which gives the loudness contour.	mean, std, min, max, median, IQR, skew, kurtosis	8
Spectral flux	Euclidean norm of the frame-to-frame difference of the magnitude spectrum. It measures how fast the spectrum is changing.	mean, std, min, max, median, IQR, skew, kurtosis	8
Spectral entropy	Shannon entropy of the power spectrum normalised to its own bin count (Eq. 1). It is a scale-free measure of how evenly energy is distributed across frequency.	mean, std, min, max, median, IQR, skew, kurtosis	8
Spectral slope	Least-squares slope of magnitude regressed on frequency, per frame. This is the spectral tilt.	mean, std, min, max, median, IQR, skew, kurtosis	8
Band-energy ratios (8 bands)	Fraction of frame energy in each of 0–500, 500–1000, 1000–2000, 2000–3000, 3000–4000, 4000–5000, 5000–6000, 6000–8000 Hz. Energy-normalised, so these describe spectral shape rather than level.	mean, std	16
High-frequency survival ratios	Fraction of frame energy above 4 and above 6 kHz. The most direct probes of lossy-codec low-pass behaviour in the inventory.	mean, std	4
Spectral contrast (7 sub-bands)	Per-band difference between spectral peaks and valleys in decibels. Band 6 covers 6–8 kHz and is the single most codec-displaced column in the table (Section III-C).	mean, std	14

the four tables is the node-level comparison summarised in Table V and Fig. 6.

TABLE XII
Cepstral descriptors. 240 columns of 436.

Descriptor	Definition	Statistics	n
ΔMFCC (20)	First-order temporal derivative of each MFCC, over a 9-frame window.	mean, std	40
ΔΔMFCC (20)	Second-order temporal derivative of each MFCC.	mean, std	40
MFCC (20)	Discrete cosine transform of the log mel-filterbank energies, which is the standard SER front end. MFCC-0 tracks overall log energy, so its derivatives measure how fast loudness changes.	mean, std, min, max, median, IQR, skew, kurtosis	160

TABLE XIII
Prosodic and voice-quality descriptors (Praat). 56 columns of 436.

Descriptor	Definition	Statistics	n
F_0 slope (mean $\lvert\cdot\rvert$)	Mean absolute first difference of the voiced F_0 contour, per second. It is the rate of pitch movement.	scalar	1
Fundamental frequency F_0	Praat autocorrelation pitch over voiced frames only.	mean, std, min, max, median, IQR, skew, kurtosis	8
Jitter (4 variants)	Cycle-to-cycle variation in glottal period: local, RAP, PPQ5, DDP. A perturbation measure of the voice source.	scalar	4
Shimmer (6 variants)	Cycle-to-cycle variation in glottal amplitude: local, local dB, APQ3, APQ5, APQ11, DDA. APQ11 needs eleven consecutive periods and is the one column in the table with non-trivial missingness (11.3 %).	scalar	6
Harmonics-to-noise ratio	Ratio of periodic to aperiodic energy in decibels. It correlates with hoarseness and breathiness.	mean, std, min, max, median	5
Formant bandwidths B_1–B_3	Median −3 dB bandwidth of the first three formants (Burg estimator).	median	1
Formants F_1–F_5	Burg-estimator formant frequencies, which describe the vocal-tract filter. F_2 movement is the strongest single correlate of disgust in this corpus.	mean, std, median	17
Intensity	Praat short-term intensity contour in decibels. Its standard deviation is among the top-ranked features for every model family in the study.	mean, std, min, max, median, IQR, skew, kurtosis	8
Voiced fraction	Proportion of frames with a defined F_0.	scalar	1
Pause fraction	Complement of the voiced fraction. By definition it equals 1− voiced fraction ($r = -1.00$).	scalar	1
Voiced-segment count	Number of maximal runs of voiced frames.	scalar	1
Voiced-segment rate	Voiced segments per second, which is a proxy for speech rate.	scalar	1
Mean voiced-segment duration	Mean length in seconds of a voiced run.	scalar	1
CPPS	Smoothed cepstral peak prominence, which is the standard acoustic correlate of breathy phonation. Drives both sadness and neutrality in every attribution method applied here.	scalar	1

TABLE XIV
Chroma descriptors. 24 columns of 436.

Descriptor	Definition	Statistics	n
Chroma (12 pitch classes)	Energy folded onto the twelve semitone pitch classes. Tuning is pinned to 0 rather than estimated, because an estimated tuning could itself shift under compression and become part of the effect being measured.	mean, std	24

TABLE XV
Global descriptors. 2 columns of 436.

Descriptor	Definition	Statistics	n
Clip duration	Decoded duration in seconds. Rises by 32.68 ms under AAC encoder priming, uniformly across every clip.	scalar	1
Peak amplitude	Maximum absolute sample value after loudness normalisation.	scalar	1

TABLE XVI
Spectral descriptors: all 114 column names.

spectral__centroid__mean	spectral__rolloff95__iqr	spectral__flux__min	spectral__band__2000__3000__std
spectral__centroid__std	spectral__rolloff95__skew	spectral__flux__max	spectral__band__3000__4000__mean
spectral__centroid__min	spectral__rolloff95__kurt	spectral__flux__median	spectral__band__3000__4000__std
spectral__centroid__max	spectral__flatness__mean	spectral__flux__iqr	spectral__band__4000__5000__mean
spectral__centroid__median	spectral__flatness__std	spectral__flux__skew	spectral__band__4000__5000__std
spectral__centroid__iqr	spectral__flatness__min	spectral__flux__kurt	spectral__band__5000__6000__mean
spectral__centroid__skew	spectral__flatness__max	spectral__entropy__mean	spectral__band__5000__6000__std
spectral__centroid__kurt	spectral__flatness__median	spectral__entropy__std	spectral__band__6000__8000__mean
spectral__bandwidth__mean	spectral__flatness__iqr	spectral__entropy__min	spectral__band__6000__8000__std
spectral__bandwidth__std	spectral__flatness__skew	spectral__entropy__max	spectral__hf_ratio__4000__mean
spectral__bandwidth__min	spectral__flatness__kurt	spectral__entropy__median	spectral__hf_ratio__4000__std
spectral__bandwidth__max	spectral__zcr__mean	spectral__entropy__iqr	spectral__hf_ratio__6000__mean
spectral__bandwidth__median	spectral__zcr__std	spectral__entropy__skew	spectral__hf_ratio__6000__std
spectral__bandwidth__iqr	spectral__zcr__min	spectral__entropy__kurt	spectral__contrast__0__mean
spectral__bandwidth__skew	spectral__zcr__max	spectral__slope__mean	spectral__contrast__0__std
spectral__bandwidth__kurt	spectral__zcr__median	spectral__slope__std	spectral__contrast__1__mean
spectral__rolloff85__mean	spectral__zcr__iqr	spectral__slope__min	spectral__contrast__1__std
spectral__rolloff85__std	spectral__zcr__skew	spectral__slope__max	spectral__contrast__2__mean
spectral__rolloff85__min	spectral__zcr__kurt	spectral__slope__median	spectral__contrast__2__std
spectral__rolloff85__max	spectral__rms__mean	spectral__slope__iqr	spectral__contrast__3__mean
spectral__rolloff85__median	spectral__rms__std	spectral__slope__skew	spectral__contrast__3__std
spectral__rolloff85__iqr	spectral__rms__min	spectral__slope__kurt	spectral__contrast__4__mean
spectral__rolloff85__skew	spectral__rms__max	spectral__band__0__500__mean	spectral__contrast__4__std
spectral__rolloff85__kurt	spectral__rms__median	spectral__band__0__500__std	spectral__contrast__5__mean
spectral__rolloff95__mean	spectral__rms__iqr	spectral__band__500__1000__mean	spectral__contrast__5__std
spectral__rolloff95__std	spectral__rms__skew	spectral__band__500__1000__std	spectral__contrast__6__mean
spectral__rolloff95__min	spectral__rms__kurt	spectral__band__1000__2000__mean	spectral__contrast__6__std
spectral__rolloff95__max	spectral__flux__mean	spectral__band__1000__2000__std	
spectral__rolloff95__median	spectral__flux__std	spectral__band__2000__3000__mean	

TABLE XVII
Cepstral descriptors: all 240 column names.

mfcc__00__mean	mfcc__07__median	mfcc__15__mean	mfcc__d1__05__mean
mfcc__00__std	mfcc__07__iqr	mfcc__15__std	mfcc__d1__05__std
mfcc__00__min	mfcc__07__skew	mfcc__15__min	mfcc__d2__05__mean
mfcc__00__max	mfcc__07__kurt	mfcc__15__max	mfcc__d2__05__std
mfcc__00__median	mfcc__08__mean	mfcc__15__median	mfcc__d1__06__mean
mfcc__00__iqr	mfcc__08__std	mfcc__15__iqr	mfcc__d1__06__std
mfcc__00__skew	mfcc__08__min	mfcc__15__skew	mfcc__d2__06__mean
mfcc__00__kurt	mfcc__08__max	mfcc__15__kurt	mfcc__d2__06__std
mfcc__01__mean	mfcc__08__median	mfcc__16__mean	mfcc__d1__07__mean
mfcc__01__std	mfcc__08__iqr	mfcc__16__std	mfcc__d1__07__std
mfcc__01__min	mfcc__08__skew	mfcc__16__min	mfcc__d2__07__mean
mfcc__01__max	mfcc__08__kurt	mfcc__16__max	mfcc__d2__07__std
mfcc__01__median	mfcc__09__mean	mfcc__16__median	mfcc__d1__08__mean
mfcc__01__iqr	mfcc__09__std	mfcc__16__iqr	mfcc__d1__08__std
mfcc__01__skew	mfcc__09__min	mfcc__16__skew	mfcc__d2__08__mean
mfcc__01__kurt	mfcc__09__max	mfcc__16__kurt	mfcc__d2__08__std
mfcc__02__mean	mfcc__09__median	mfcc__17__mean	mfcc__d1__09__mean
mfcc__02__std	mfcc__09__iqr	mfcc__17__std	mfcc__d1__09__std
mfcc__02__min	mfcc__09__skew	mfcc__17__min	mfcc__d2__09__mean
mfcc__02__max	mfcc__09__kurt	mfcc__17__max	mfcc__d2__09__std
mfcc__02__median	mfcc__10__mean	mfcc__17__median	mfcc__d1__10__mean
mfcc__02__iqr	mfcc__10__std	mfcc__17__iqr	mfcc__d1__10__std
mfcc__02__skew	mfcc__10__min	mfcc__17__skew	mfcc__d2__10__mean
mfcc__02__kurt	mfcc__10__max	mfcc__17__kurt	mfcc__d2__10__std
mfcc__03__mean	mfcc__10__median	mfcc__18__mean	mfcc__d1__11__mean
mfcc__03__std	mfcc__10__iqr	mfcc__18__std	mfcc__d1__11__std
mfcc__03__min	mfcc__10__skew	mfcc__18__min	mfcc__d2__11__mean
mfcc__03__max	mfcc__10__kurt	mfcc__18__max	mfcc__d2__11__std
mfcc__03__median	mfcc__11__mean	mfcc__18__median	mfcc__d1__12__mean
mfcc__03__iqr	mfcc__11__std	mfcc__18__iqr	mfcc__d1__12__std
mfcc__03__skew	mfcc__11__min	mfcc__18__skew	mfcc__d2__12__mean
mfcc__03__kurt	mfcc__11__max	mfcc__18__kurt	mfcc__d2__12__std
mfcc__04__mean	mfcc__11__median	mfcc__19__mean	mfcc__d1__13__mean
mfcc__04__std	mfcc__11__iqr	mfcc__19__std	mfcc__d1__13__std
mfcc__04__min	mfcc__11__skew	mfcc__19__min	mfcc__d2__13__mean
mfcc__04__max	mfcc__11__kurt	mfcc__19__max	mfcc__d2__13__std
mfcc__04__median	mfcc__12__mean	mfcc__19__median	mfcc__d1__14__mean
mfcc__04__iqr	mfcc__12__std	mfcc__19__iqr	mfcc__d1__14__std
mfcc__04__skew	mfcc__12__min	mfcc__19__skew	mfcc__d2__14__mean
mfcc__04__kurt	mfcc__12__max	mfcc__19__kurt	mfcc__d2__14__std
mfcc__05__mean	mfcc__12__median	mfcc__d1__00__mean	mfcc__d1__15__mean
mfcc__05__std	mfcc__12__iqr	mfcc__d1__00__std	mfcc__d1__15__std
mfcc__05__min	mfcc__12__skew	mfcc__d2__00__mean	mfcc__d2__15__mean
mfcc__05__max	mfcc__12__kurt	mfcc__d2__00__std	mfcc__d2__15__std
mfcc__05__median	mfcc__13__mean	mfcc__d1__01__mean	mfcc__d1__16__mean
mfcc__05__iqr	mfcc__13__std	mfcc__d1__01__std	mfcc__d1__16__std
mfcc__05__skew	mfcc__13__min	mfcc__d2__01__mean	mfcc__d2__16__mean
mfcc__05__kurt	mfcc__13__max	mfcc__d2__01__std	mfcc__d2__16__std
mfcc__06__mean	mfcc__13__median	mfcc__d1__02__mean	mfcc__d1__17__mean
mfcc__06__std	mfcc__13__iqr	mfcc__d1__02__std	mfcc__d1__17__std
mfcc__06__min	mfcc__13__skew	mfcc__d2__02__mean	mfcc__d2__17__mean
mfcc__06__max	mfcc__13__kurt	mfcc__d2__02__std	mfcc__d2__17__std
mfcc__06__median	mfcc__14__mean	mfcc__d1__03__mean	mfcc__d1__18__mean
mfcc__06__iqr	mfcc__14__std	mfcc__d1__03__std	mfcc__d1__18__std
mfcc__06__skew	mfcc__14__min	mfcc__d2__03__mean	mfcc__d2__18__mean
mfcc__06__kurt	mfcc__14__max	mfcc__d2__03__std	mfcc__d2__18__std
mfcc__07__mean	mfcc__14__median	mfcc__d1__04__mean	mfcc__d1__19__mean
mfcc__07__std	mfcc__14__iqr	mfcc__d1__04__std	mfcc__d1__19__std
mfcc__07__min	mfcc__14__skew	mfcc__d2__04__mean	mfcc__d2__19__mean
mfcc__07__max	mfcc__14__kurt	mfcc__d2__04__std	mfcc__d2__19__std

TABLE XVIII
Prosodic and voice-quality descriptors (Praat): all 56 column names.

prosody__f0__mean	prosody__jitter__local	prosody__hnr__median	prosody__f4__std
prosody__f0__std	prosody__jitter__rap	prosody__f1__mean	prosody__f4__median
prosody__f0__min	prosody__jitter__ppq5	prosody__f1__std	prosody__f5__mean
prosody__f0__max	prosody__jitter__ddp	prosody__f1__median	prosody__f5__std
prosody__f0__median	prosody__shimmer__local	prosody__f1__bw__median	prosody__f5__median
prosody__f0__iqr	prosody__shimmer__localdb	prosody__f2__mean	prosody__intensity__mean
prosody__f0__skew	prosody__shimmer__apq3	prosody__f2__std	prosody__intensity__std
prosody__f0__kurt	prosody__shimmer__apq5	prosody__f2__median	prosody__intensity__min
prosody__voiced__fraction	prosody__shimmer__apq11	prosody__f2__bw__median	prosody__intensity__max
prosody__f0__slope__abs__mean	prosody__shimmer__dda	prosody__f3__mean	prosody__intensity__median
prosody__n__voiced__segments	prosody__hnr__mean	prosody__f3__std	prosody__intensity__iqr
prosody__voiced__rate__per__s	prosody__hnr__std	prosody__f3__median	prosody__intensity__skew
prosody__mean__voiced__seg__s	prosody__hnr__min	prosody__f3__bw__median	prosody__intensity__kurt
prosody__pause__fraction	prosody__hnr__max	prosody__f4__mean	prosody__cpps

TABLE XIX
Chroma descriptors: all 24 column names.

chroma__00__mean	chroma__04__mean	chroma__08__mean
chroma__00__std	chroma__04__std	chroma__08__std
chroma__01__mean	chroma__05__mean	chroma__09__mean
chroma__01__std	chroma__05__std	chroma__09__std
chroma__02__mean	chroma__06__mean	chroma__10__mean
chroma__02__std	chroma__06__std	chroma__10__std
chroma__03__mean	chroma__07__mean	chroma__11__mean
chroma__03__std	chroma__07__std	chroma__11__std

TABLE XX
Global descriptors: all 2 column names.

duration__s	peak__amplitude

TABLE XXI
Entropy tree fitted on Uncompressed WAV, all internal nodes to depth 4.

Path	Feature	Threshold	H	IG	n
root	prosody__intensity__std	8.2184	2.5850	0.3090	4905
L	mfcc__d1__00__std	3.9965	2.3849	0.1415	3065
R	mfcc__d1__00__std	6.6630	2.0891	0.1797	1840
LL	spectral__band__0__500__mean	0.8626	1.9382	0.0685	1059
LR	prosody__f0__mean	220.5886	2.3995	0.0990	2006
RL	prosody__f0__median	161.8673	2.2633	0.1141	847
RR	prosody__intensity__min	42.7554	1.6055	0.0925	993
LLL	prosody__intensity__median	64.3901	2.1110	0.0763	503
LLR	prosody__f0__mean	174.4501	1.6481	0.0727	556
LRL	spectral__centroid__iqr	209.5395	2.3328	0.0682	1781
LRR	prosody__f0__max	372.4767	2.0340	0.1797	225
RLL	spectral__centroid__iqr	310.5503	2.3335	0.1579	254
RLR	prosody__f0__min	149.0330	2.0683	0.0778	593
RRL	mfcc__03__max	46.1135	1.2265	0.0747	542
RRR	spectral__contrast__2__mean	12.1160	1.8550	0.0915	451
LLLL	mfcc__05__iqr	8.7058	2.3273	0.1863	139
LLLR	mfcc__07__mean	1.0004	1.9225	0.0759	364
LLRL	mfcc__00__mean	-220.3358	1.6344	0.0676	391
LLRR	prosody__intensity__iqr	5.8890	1.4344	0.1343	165
LRLL	mfcc__d1__03__std	1.3160	2.2210	0.0810	564
LRLR	duration__s	2.6526	2.2848	0.0690	1217
LRRL	mfcc__d1__00__std	5.5015	2.2359	0.2615	119
LRRR	mfcc__d2__05__std	0.6661	1.4206	0.2061	106
RLLL	spectral__contrast__1__std	3.1735	2.2781	0.1619	146
RLLR	duration__s	2.4524	2.0339	0.1987	108
RLRL	mfcc__03__std	12.0396	2.1028	0.0993	385
RLRR	prosody__intensity__std	10.4187	1.7818	0.1437	208
RRLL	spectral__entropy__mean	0.4682	1.5107	0.1241	305
RRLR	mfcc__d1__00__std	8.0503	0.6898	0.0951	237
RRRL	mfcc__d2__00__std	3.2530	2.1332	0.2152	95
RRRR	mfcc__03__max	40.8027	1.6633	0.1040	356

TABLE XXII
Entropy tree fitted on MP3 @ 64 kbit/s, all internal nodes to depth 4.

Path	Feature	Threshold	H	IG	n
root	prosody__intensity__std	8.2210	2.5850	0.3090	4905
L	mfcc__d1__00__std	4.0175	2.3849	0.1398	3065
R	mfcc__d1__00__std	6.6883	2.0891	0.1800	1840
LL	spectral__band__0__500__mean	0.8627	1.9524	0.0669	1080
LR	prosody__f0__mean	221.5304	2.3997	0.1002	1985
RL	prosody__f0__median	161.8687	2.2615	0.1129	873
RR	prosody__intensity__min	42.3291	1.5890	0.0914	967
LLL	prosody__intensity__median	63.9287	2.1257	0.0752	515
LLR	prosody__f0__mean	175.0993	1.6634	0.0762	565
LRL	spectral__band__0__500__std	0.2524	2.3332	0.0688	1771
LRR	mfcc__02__std	14.7187	2.0072	0.1668	214
RLL	spectral__centroid__iqr	310.5299	2.3306	0.1592	257
RLR	spectral__band__0__500__mean	0.7316	2.0707	0.0767	616
RRL	spectral__band__3000__4000__mean	0.0146	1.2123	0.0774	532
RRR	spectral__contrast__2__mean	12.2081	1.8440	0.1046	435
LLLL	mfcc__05__iqr	8.7206	2.3261	0.1391	148
LLLR	spectral__flatness__mean	0.0041	1.9388	0.0744	367
LLRL	mfcc__00__mean	-223.6980	1.6513	0.0753	399
LLRR	prosody__intensity__iqr	5.8541	1.4319	0.1347	166
LRLL	prosody__hnr__median	3.7622	2.2810	0.0703	935
LRLR	mfcc__d2__00__std	2.4823	2.2461	0.0777	836
LRRL	mfcc__06__mean	-1.1106	1.4472	0.3107	65
LRRR	mfcc__16__std	4.4168	2.0110	0.1969	149
RLLL	mfcc__05__median	2.2367	2.2728	0.1562	147
RLLR	duration__s	2.4524	2.0326	0.2006	110
RLRL	prosody__f0__median	270.2333	2.0442	0.1015	320
RLRR	prosody__f0__std	46.8924	1.9399	0.1148	296
RRLL	mfcc__03__max	39.2929	1.4878	0.1099	272
RRLR	mfcc__d2__10__std	0.6478	0.7657	0.0986	260
RRRL	mfcc__d2__00__std	3.2140	2.1218	0.2554	106
RRRR	mfcc__d2__12__std	0.5605	1.6136	0.0943	329

TABLE XXIII
Entropy tree fitted on MP4/AAC @ 64 kbit/s, all internal nodes to depth 4.

Path	Feature	Threshold	H	IG	n
root	prosody__intensity__std	8.3241	2.5850	0.3064	4905
L	mfcc__00__iqr	60.5853	2.3887	0.1408	3075
R	mfcc__d1__00__std	7.1056	2.0879	0.1555	1830
LL	spectral__band__500__1000__mean	0.0797	1.9819	0.0875	1181
LR	prosody__f0__mean	219.0372	2.4096	0.0957	1894
RL	prosody__f0__median	161.0991	2.2673	0.1047	874
RR	prosody__intensity__std	11.2496	1.6245	0.0884	956
LLL	prosody__f0__mean	169.4067	1.4798	0.0877	395
LLR	prosody__f3__bw__median	712.6446	2.0975	0.0757	786
LRL	spectral__band__0__500__mean	0.8284	2.3465	0.0696	1662
LRR	mfcc__02__std	14.7880	2.0711	0.1528	232
RLL	spectral__band__1000__2000__std	0.0765	2.3608	0.1810	226
RLR	spectral__band__0__500__mean	0.7133	2.0917	0.0718	648
RRL	spectral__contrast__2__mean	12.2765	1.8882	0.1074	461
RRR	mfcc__03__max	38.0092	1.2059	0.0851	495
LLLL	mfcc__05__iqr	7.6452	1.3054	0.1059	241
LLLR	prosody__f2__std	326.8660	1.5276	0.1362	154
LLRL	prosody__f0__mean	199.2747	2.1863	0.1157	498
LLRR	spectral__contrast__1__mean	7.7606	1.7295	0.0923	288
LRLL	duration__s	2.8480	2.2806	0.0690	1204
LRLR	prosody__cpps	5.3755	2.2671	0.0731	458
LRRL	mfcc__13__kurt	0.3511	1.5158	0.2506	66
LRRR	spectral__band__5000__6000__std	0.0311	2.0779	0.1904	166
RLLL	mfcc__00__iqr	112.3155	2.2882	0.1462	170
RLLR	spectral__band__500__1000__mean	0.1727	1.8385	0.2681	56
RLRL	prosody__f0__median	270.2040	2.0207	0.1057	296
RLRR	prosody__f0__slope__abs__mean	704.7569	2.0192	0.1053	352
RRLL	mfcc__05__median	3.1128	2.1051	0.1682	126
RRLR	spectral__band__0__500__std	0.2993	1.6568	0.1037	335
RRRL	prosody__f4__median	3950.6527	1.5464	0.1841	122
RRRR	spectral__band__3000__4000__mean	0.0143	0.9815	0.0752	373

TABLE XXIV
Entropy tree fitted on Roundtrip WAV (AAC re-wrapped), all internal nodes to depth 4.

Path	Feature	Threshold	H	IG	n
root	prosody_intensity_std	8.3241	2.5850	0.3064	4905
L	mfcc_00_iqr	60.5802	2.3887	0.1408	3075
R	mfcc_d1_00_std	7.1058	2.0879	0.1555	1830
LL	spectral_band_500_1000_mean	0.0782	1.9819	0.0882	1181
LR	prosody_f0_mean	219.0372	2.4096	0.0957	1894
RL	prosody_f0_median	161.0993	2.2673	0.1047	874
RR	prosody_intensity_std	11.2497	1.6245	0.0884	956
LLL	prosody_f0_mean	170.6874	1.4582	0.0902	381
LLR	prosody_f3_bw_median	712.5267	2.0961	0.0727	800
LRL	spectral_band_0_500_mean	0.8280	2.3465	0.0701	1662
LRR	mfcc_02_std	14.7881	2.0711	0.1528	232
RLL	spectral_band_1000_2000_std	0.0765	2.3608	0.1810	226
RLR	spectral_band_0_500_mean	0.7136	2.0917	0.0728	648
RRL	spectral_contrast_2_mean	12.2826	1.8882	0.1051	461
RRR	mfcc_03_max	42.9480	1.2059	0.0846	495
LLLL	mfcc_02_min	-4.2720	1.3393	0.0935	239
LLLR	spectral_flatness_median	0.0075	1.4163	0.1184	142
LLRL	prosody_f0_mean	199.2748	2.1897	0.1143	505
LLRR	spectral_contrast_1_mean	7.7596	1.7311	0.0940	295
LRLL	duration_s	2.8480	2.2798	0.0680	1203
LRLR	prosody_f1_bw_median	281.4372	2.2671	0.0744	459
LRRL	mfcc_13_kurt	0.3489	1.5158	0.2506	66
LRRR	spectral_band_5000_6000_std	0.0311	2.0779	0.1904	166
RLLL	mfcc_00_iqr	112.3050	2.2882	0.1462	170
RLLR	spectral_band_500_1000_mean	0.1727	1.8385	0.2681	56
RLRL	prosody_f0_median	270.2047	2.0178	0.1036	299
RLRR	prosody_f0_slope_abs_mean	704.7769	2.0200	0.1026	349
RRLL	spectral_hf_ratio_4000_std	0.0443	2.1003	0.1608	128
RRLR	spectral_band_0_500_std	0.3006	1.6592	0.1006	333
RRRL	spectral_band_0_500_mean	0.6868	1.5345	0.1361	210
RRRR	spectral_band_3000_4000_mean	0.0144	0.8168	0.1042	285